Analysing the Suitability of IEEE 802.11ah for Next Generation Internet of Things: A Comparative Study

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Abstract

Hyperconnectivity, real-time processing, and interoperable communication facilities are key to the Next Generation (NG) Internet of Things (IoT). However, existing IoT architectures employ a diverse range of communication technologies to support desired data rates and coverage range, making them more complex to manage. The emerging IEEE 802.11ah standard, marketed as WiFi HaLow^{TM}, enables seamless communication among large-scale and energy-efficient sensor/actuator nodes to the Internet. IEEE 802.11ah introduces key features like Restricted Access Window (RAW) for reducing contention, Hierarchical Traffic Indication Map (TIM) and Association Identifier (AID) for efficient scheduling and device identification, and multiple Modulation and Coding Schemes (MCSs) for adaptive data transmission, enhancing IoT connectivity in Next-Generation IoT (NGI) communication. In this paper, we discuss IEEE 802.11ah for NGI concerning different requirements in terms of communication and computation. We survey the suitability of 802.11ah for NGI as compared to the existing state-of-the-art connectivity solutions. Thereafter, an IEEE 802.11ah-based NGI architecture and protocol stack are proposed and discussed, according to the different distinct features of this novel technology. Finally, we conclude by enumerating research and future directions for the proposed network architecture.

Keywords: Internet of Things (IoT), Next Generation IoT, Wi-Fi HaLow, IEEE 802.11ah, sub-1 GHz, IDN

1. Introduction

The Internet of Things (IoT) as a technology is experiencing widespread adoption owing to the rapid advancements in machine-to-machine communication technologies. This progress enables the seamless integration of billions of objects from the physical environment to the Internet, including sensors, actuators, Radio-Frequency IDentification (RFID) devices, and more [1, 2]. These objects (i.e., Things in IoT) are embedded devices with varied communica-

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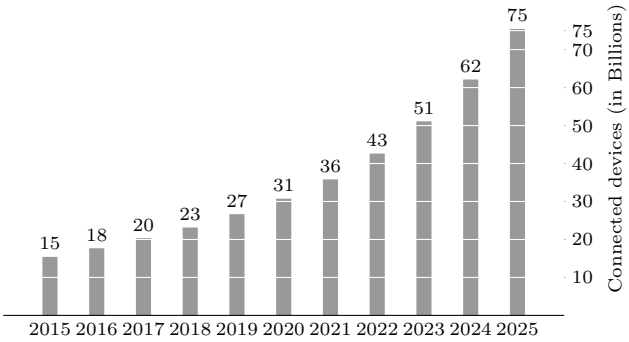


Figure 1: Number of IoT connected devices installed base worldwide from 2015 to 2025 with forecasts from 2022 to 2030 (in billions) [4]

tion capabilities and can connect to both local and central servers[1]. The projected number of connected devices worldwide for the period 2015 to 2025 is shown in Fig. 1. Wireless network technologies or communication facilities are the solutions for connecting such a massive quantity of devices without bothering about their geographical location [3]. For enabling holistic and seamless IoT operations, it requires a suitable wireless communication standard. Considering a complete network solution for IoT applications such as healthcare, agriculture, vehicular, and industry requires different communication technologies for sensor and back-haul scenarios.

As IoT emerged as an add-on to the existing Internet, it is important to enhance this architecture by integrating the add-on for Next-Generation IoT [5]. We use the term Next Generation Internet of Things and NGI interchangeably in the rest of the paper. To provide wide area coverage and low bandwidth services to IoT, already established technologies such as IEEE 802.15.4, Bluetooth Low Energy (BLE), or RFID could only be able to provide ranges from 1 to 100 meters area as well as bandwidth from bytes to megabytes count. Moreover, these technologies are yet in need of advancement in energy efficiency for a durable network existence. As a solution to this, 100 meters to a few kilometres of service enabled by technologies such as Wireless Fidelity (WiFi), Worldwide Interoperability for Microwave Access (WiMAX), General Packet Radio Services (GPRS), and Long-Term Evolution (LTE). These technologies are created with Access Point (AP) based architectures demanding proper line of sights [6]. Over the advantages of gaining high throughput (kilobytes to gigabytes) by these technologies, yet they face a major challenge of high energy consumption. Also, they face challenges while making the network functional with large-scale heterogeneous devices in various conditions. A limited number of devices, small area coverage, and low throughput are the main constraints of these technologies, resulting in less scalable network architecture. Similarly, Low-Power Wide-Area Network (LPWAN) technologies such as LoRa [7], SigFox [8] have gained lots of interest as candidate technologies for IoT. However, they support a more extended coverage range, but it is not befitting many IoT applications due to their low data rate. On the other hand, the recently proposed WiFi 6 [9] has specifically

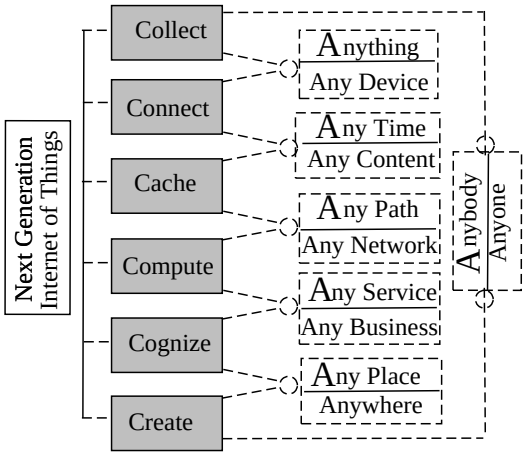


Figure 2: The next generation IoT hyper-connected: the 6A's and 6C's [5]

worked to provide high throughput. Hence, it is difficult to manage such a complex IoT network. IEEE 802.11ah standard eliminates the above-mentioned research lacuna up to a reasonable extent using its new PHY and MAC layer supports.

Fig. 2 shows different requirements of NGI [5] showing the abstract view. For these, a wireless network solution that can seamlessly connect all the network components concerning the resource-constraint nature of the devices. IEEE 802.11ah is expected to connect such devices and reducing the management and power-saving mechanism. The NGI with 802.11ah has the potential to connect and process devices and data in the IoT seamlessly.

1.1. Motivation: IEEE 802.11ah for IoT for reducing complexity and co-existence

As the demand for integration of a large number of devices in IoT networks is tremendously increasing, we look forward to picking up resilient communication technology facilitating scalability. In the existing infrastructure, heterogeneous technologies are clubbed together, which creates more complexities. For example, the existing real-life IoT testbeds, as presented in Table 1, uses a combination of different technologies and protocols to fulfil the requirements of the applications. Smart lighting, smart grid, smart environment, smart healthcare and multiple such smart applications individually deploy thousands of devices, and all of these are part of a single network eco-system[10]. Traffic patterns in such a network are also not the same. In smart parking applications, based on the arrival and departure events created by a car, the packet generation rate varies [11]. Monitoring applications like smart environment monitoring and smart grid mostly generate traffic periodically. These devices are restricted by—*processing, storing*, and *energy saving* capabilities. These applications use different wireless standards considering their known benefits. As a result, many heterogeneous wireless networks are competing for the same frequency in the same

Table 1: Comparison of selected real-life deployment IoT testbeds

Testbed	**Description**	**Technologies used**	**Targeted applications**	**Large area**	**Large scale**	**Scalability**	**Heterogeneity**	**Low cost**	**Low power**
USN [13]	An infrastructure for sensor networks in urban areas	ZigBee, 6LoWPAN, ISA-100.11.a, xDSL, GPRS, etc.	Urban IoT and Smart city	✗	✗	✓	✗	✓	✗
BUCL [14]	An smart city testbed for Birmingham city in England	IEEE 802.11b/g	Smart environment monitoring in urban areas	✓	✓	✓	✓	✓	✓
SmartSantander [15]	An IoT-based Smart city network deployed over Santander city of Spain	GPRS, IEEE 802.15.4, NFC, RFID	Smart lighting, Smart parking, transportation etc.	✓	✗	✗	✗	✗	✗
JEJU Testbed [16]	Testbeds setup for both residential and commercial spaces at JEJU Island in Korea	ZigBee, WiFi, PLC, WiMAX, CDMA, WCDMA	Smart grid for solar and wind power monitoring	✓	✗	✗	✗	✗	✗
SmartGridLab [17]	A laboratory environment for research purpose	IEEE 802.15.4, ISA100.11a, WirelessHART, MiWi	Smart grid	✗	✗	✗	✗	✗	
EAR-IT [18]	A laboratory environment for large-scale IoT	IEEE 802.15.4 based communication technologies	Smart building, Smart city	✗	✗	✗	✗	✗	✗
OpenTestBed [19]	An open testbed for IoT	IEEE 802.11a/b/g/n/ac, LTE, IEEE 802.15.4	Performance analysis of IoT protocols	✗	✗	✗	✗	✗	✗

coverage area [12]. Again, the chances of interference due to the coexistence of multiple types of networks are also high. The table provides a detailed comparison of selected real-life deployment IoT testbeds, offering insights into their key characteristics and capabilities. Notably, the absence of certain features in specific testbeds highlights the diversity and unique focuses of each environment. For instance, the USN testbed considers a small area coverage, indicating its more localized sensor network infrastructure tailored for urban areas. Similarly, SmartSantander lacks large-scale deployment capabilities, suggesting that its primary applications, such as smart lighting and parking, are concentrated in specific areas rather than spanning across a wide geographical area. The testbeds EAR-IT, SmartGridLab, and OpenTestBed exhibit a comprehensive set of features unavailability, implying limitations in scalability, heterogeneity, low cost, and low power attributes. This nuanced analysis of unavailability underscores the varying objectives and specializations of each IoT testbed, providing valuable insights for researchers and practitioners in selecting an appropriate platform based on their specific requirements.

Table 2: Key features and example uses-cases of IEEE 802.11ah

Applications	**Traffic flow**	**Data rate**	**Coverage**	**References**
Smart grid	Bi-Directional	100Kbps	100m-3Km	[23, 24, 25, 26, 17]
Environment/Agriculture Monitoring	Mostly Uplink	100Kbps	1Km-2Km	[27, 22]
Industrial process automation	Bi-Directional	<1Mbps	500m-1km	[28, 29]
Indoor healthcare system	Mostly Uplink	100Kbps	30-100m	[30]
Remote Healthcare/Fitness	Bi-Directional	100Kbps	500m-1km	[31, 32]
Backhaul/Backbone	Bi-Directional	>200Kbps	1km-3km	[33, 34]

IEEE 802.11ah offers a solution to address the problem of scalability, and it also supports heterogeneous application types using licence-free channel band (Sub-1 GHz) and multiple modulation schemes (MCS0-MCS9). A very high data rate (150kbps-78mbps) service by 802.11ah technology is a desirable feature for IoT devices with limited energy ([20], and [21]).

When devices are awake for transmission, they constantly consume energy, which is a crucial factor for energy-constrained IoT devices. When the devices do not frequently transmit, they are set to be awake for a specific time slot to save their energy. Table 2, we detailed about different traffic requirements associated with various applications. IEEE 802.11ah technology gives coexistence management and flexible service to numerous large-scale IoT applications previously mentioned with reduced computation and communication cost. In this case, comparing to IEEE 802.15.4, technology is insufficient to prove coexistence management due to its 2.4GHz channel band [22]. Moreover, the existing solution fails when the WiFi network needs to cover a wide area maintaining high bandwidth across the service area. For instance, if we consider a smart agriculture application, where our task is to monitor the cultivation remotely and get alerted for any damages when required. In this case, the damages may happen due to forest fire or wildlife; the WiFi must have large area connectivity to detect these damages. While monitoring the region, our data is of image type, and this requires high throughput. IEEE 802.11ah enables its facility in such a scenario with its very long-distance connectivity service resulting in economic loss and fire detection. In another instance of smart-grid, to monitor smart meters and protect intelligent devices from damage, 802.11ah is a reliable solution [17]. Network complexity, coexistence, and management are the challenges facing by existing heterogeneous wireless network architectures. The relay-based architecture of IEEE 802.11ah can serve a range of 1km radius with single-hop connectivity, provisioning a maximum of 2-hops. Other than sensing and actuating tasks, WiFi-HaLoW can investigate the vulnerabilities and run telemetry in different applications such as industry.

1.2. Architecture of Next-Generation IoT

Adding new technologies such as Artificial Intelligence and advanced communication networks to the advancement of the IoT architecture to meet users' needs for quality of service (QoS), performance, and authentication [35]. The

NGI devices, intelligent edge computing, a robust and reliable infrastructure, and a data-centric architecture play game-changer rules in this advanced IoT architecture. Interactive smart IoT devices are arising rapidly to offer communication between humans and things. And to take this to the advanced level, edge computing is evolved to bring forth the services such as real-time computation, control, context-aware, etc. Edge computing architecture supports decentralized decision-making capacity at the network edge devices [36]. In the Edge computing paradigm, as the computation is done at the devices nearest to the user, which are network devices placed at the edge provides real-time processing of data in addition to ensuring strong security. Combining edge computing with artificial intelligence allows us to grab the benefits of smart and latency-sensitive services as intelligent edge [37]. Resource constraint IoT devices are not capable of handling gigantic data getting generated by sensor devices each day. It must deal with these data for pre-processing and computation at the Artificial Intelligence (AI) level. The AI is getting deployed at the device level through the AI application originates at the Cloud level to reduce latency and obtain service efficiency and network reliability. Addressing all the drawbacks of present IoT architecture, we are inclined towards a more reliable and resilient architecture, i.e., NGI architecture. NGI devices such as WiFi 6 are to provide a top speed of 9.6 Gbps while improving the network with piles of devices associated with it [38]. NGI is emerging with its heights of capabilities in different applications such as smart-healthcare, smart-grid, and 6G applications. IEEE 802.11ah combines WiFi and Low-Power communication technologies' benefits, providing a high data rate and different application-specific services. Moreover, this WiFi standard guarantees services such as low-power energy consumption, large-scale deployment, and higher QoS [39].

The existing works such as [5], and [40] highlight different applications and requirements of NGI. Shafique *et al.* [41] proposes a road-map for NGI while discussing machine learning-based approaches. However, connectivity and communication are the keys to the road-map of NGI. With the growing number of devices and other IoT components, network complexity and management are the concerns that should be considered in the NGI architecture.

Related works pertaining to the review scope and the exploration of potential future applications of 802.11ah have already been presented in various existing studies. In the study conducted by Tian *et al.* [42], a comprehensive analysis of WiFi HaLow was presented, focusing on its unique features at the MAC and PHY layers. The authors discussed the effective utilization of these features to meet the performance requirements of IoT applications. Additionally, the research paper emphasized the various challenges faced in IEEE 802.11ah, including issues related to heterogeneous network types, QoS, multiple access points, and coexistence within sub-1GHz frequency bands. For example, Ahmed at al. [43] present a detail discussion on MAC layer of 802.11ah while issues and challenges in the different problem domains of the IEEE 802.11ah MAC protocol and analyzes the recently proposed solutions. However, this paper not only reviews the network architecture for Next Generation IoT (NGI), but also explores the features of existing communication technologies and assesses the feasibility of integrating the IEEE 802.11ah

standard into the NGI framework. Furthermore, we evaluate the performance of 802.11ah and discuss the additional support required to create a fully capable infrastructure for NGI.

This paper introduces a novel approach to designing the architecture for the next generation of Internet of Things (IoT) by leveraging the advanced capabilities of IEEE 802.11ah. It provides an overview of the essential parameters and requirements for the next generation IoT and conducts a comparative analysis of various state-of-the-art technologies in this domain. Furthermore, the paper offers a comprehensive analysis of the current advancements in IEEE 802.11ah hardware and software. It establishes a correlation between the necessary features and the enabling components for the next generation IoT. We also discuss a few performance results of this technology. However, the scope of our performance analysis is limited to the network and communication aspects when evaluating the performance of NGI. Finally, the paper identifies and discusses several open research challenges that need to be addressed in this direction.

Table 3 presents a list of abbreviations and their corresponding descriptions, which are used repeatedly in the paper.

1.3. Contributions

This paper initially discusses the key requirements and characteristics of NGI and then proposes an IEEE 802.11ah-based NGI. We emphasize the challenges and further opportunities of 802.11ah and how to address them, as well as the aspects that may be improved while designing an NGI architecture. The following summarizes our key *contributions* in this paper comparing to recent works:

- We discuss the characteristics of communication technologies, different supported standards, and the essential requisite of an NGI architecture.
- We compare the existing state-of-the-art communication standards.
- Concerning the PHY and MAC layer features as provided by IEEE 802.11ah, this paper proposes an IEEE 802.11ah-based scalable NGI architecture.
- Compared to the other NGI architectures, this approach considers communication and connectivity solutions with recent features such as programmable, intelligence, and cognitive.
- Open research issues, challenges, and future works of the proposed architecture are provided.

1.4. Paper organization

We organized this paper into nine sections. Section 2 discusses the requirements of Next-generation IoT architecture. A detailed analysis and comparison among various existing communication technologies are given in Section 3.

Table 3: Abbreviations and Acronyms

5G	Fifth generation	6G	Sixth generation
AI	Artificial intelligence	AID	Association IDentifier
AP	Access point	BLE	Bluetooth low energy
BSS	Basic Service Set	CSB	Cross-slot boundary
D-TIM	(Delivery) Traffic indication map	EDCA	Enhanced distributed channel access
FTT	Fast Fourier Transform	GPRS	General packet radio services
IDN	Intent-driven network	IoT	Internet of Things
IP	Internet protocol	ISM	Industrial, scientific, and medical
LPWAN	Low-power wide-area network	LTE	Long-term evolution
M2M	Machine to machine	MAC	Medium access control
MCS	Modulation and coding schemes	MIMO	Multiple input multiple output
NB	Narrow band	NFC	Near field communication
NG(I)	Next generation (IoT)	OFDM	Orthogonal frequency-division multiplexing
PHY	Physical	PRR	Packet received ratio
QoS	Quality of service	RAP	Relay AP
(P)RAW	(Periodic) Restricted access window	RFID	Radio-frequency IDentification
SDN	Software defined networks	SIFS	Short interframe space
STA	Station/node	TWT	Target wait time
TXOP	Transmission opportunity	WiFi	Wireless fidelity
WiMAX	Worldwide interoperability for microwave access	6LoWPAN	IPv6 over Low-power wireless personal area networks

Features of the proposed next-generation IoT and its 802.11ah enabled architecture is provided in Section 4 and 5, respectively. We provided the IEEE 802.11ah based IoT stack in Section 6, followed by presenting a detailed performance analysis of this new architecture in Section 7. We give brief information about the current status of the IEEE 802.11ah-based hardware and software stack, and current research status and future research directions are briefly discussed in Section 8. In Section 9, we conclude this paper.

2. Requirements of Next-Generation Internet of Things

The number of communicating devices due to recent M2M communication will lead to an annual predicted growth rate greater than 25% [44]. Communication among such a large number of devices using limited channels has been a severe problem due to the lack of proper MAC and PHY design [45]. Collision possibilities in a high-density network are also very high in a high-density network. The existing literature has proposed several methods in order to combat this. As compared to traditional networking, IoT uses resource constraint devices in terms of low power consumption, low processing, and low storage capacity devices for sensing and actuation operation. Moreover, traffic patterns in such networks are also heterogeneous in terms of packet size, interval, and event. Eventually, there emerges a need for new MAC and PHY protocols, which incorporate such dynamic behaviours of a network. We discuss the key

characteristics of NGI in the following subsections.

Scalability: A communication technology should be able to control colossal contention and collisions to enable stable performances in a shared wireless medium with an increasing number of IoT devices [46]. The performance of the contention-based protocols usually degrades due to the collision in transmission. Contention-free protocol usually requires additional implementation overhead. In dynamic and heterogeneous applications the technology should effectively adapt and balance with changing node densities. For example, in a smart grid application, the implementation of a scalable smart grid infrastructure allows utility companies to efficiently manage electricity distribution across a large geographical area by integrating renewable energy sources, monitoring power consumption, and dynamically adjusting energy flow based on demand.

Energy efficiency: Enabling communication features in tiny devices, we must adopt a mechanism for energy efficient critical design parameters for IoT protocol. Factors such as the battery, economic impact, and environmental impact are the primary reasons for designing energy-efficient protocols. Information and communications technology (ICT) industries currently carrying 1.3% of total harmful outflow in the world, which is increasing with the increasing number of things in the coming times [47]. Smart homes equipped with energy management systems that optimize the use of appliances, lighting, and heating/cooling, identifying energy wastage, optimizing usage patterns, and making informed decisions to improve overall energy efficiency.

High throughput: With the increasing capacity of the PHY layer, the MAC layer scalability in terms of channel access should improve as well. Throughput efficient MAC/PHY layer solutions need to be designed for accommodating a large number of devices in a limited channel/spectrum [47]. Some applications, such as video surveillance, voice over IP, etc., require high bandwidth to run. The channel access mechanism should reduce the excess delay due to collisions and control frames. For example, high throughput is a crucial requirement in applications such as industrial automation and smart healthcare where wireless sensor networks are employed to collect real-time data from a large number of sensors and transmit it with high throughput to control systems for monitoring and decision-making.

Low overhead: A technology should minimize the protocol overhead when a massive number of devices communicate over a large area [48]. The per-packet overhead and higher time consumption on account of backoff, inter-frame spacing, and handshakes need to be relatively smaller. Different control packets like RTS, CTS, ACK, etc., handshake create a massive overhead in the network. This overhead is found to be one of the significant sources of energy wastage and may hold 40 to 75% of the total channel capacity ([49] and [50]). Low-power IoT devices with low overhead communication protocols, such as LoRaWAN or Sigfox, are widely used for tracking assets in logistics and supply chain management, providing real-time location updates while conserving battery life.

Adaptive: A technology should be able to adapt itself along with the changing nature of IoT and its traffic. It

is essential to understand the traffic patterns of different applications for better utilization of the network resources. Asynchronous traffic is time-bound in nature that must be transferred within a specific time frame, which has a low tolerance for delay and loss. In IoT, control frames are exchanged among nodes to keep the nodes alive, but the traffic generation of data packets depends on applications and scenarios. There are heterogeneous devices, various application scenarios, different data formats, and communication networks in many situations. The protocols should adaptively change their behaviour with the current conditions of the environment [2]. For example, when a person enters a room, all the devices should adaptively help the network provide better smart home services. A protocol can adapt itself to provide sufficient channels/resources and save energy for the remaining time in these cases.

Fast and reliable: MAC and PHY layer protocols to deal with the unusual but mission-critical application traffic are essential. There are many IoT applications where the delay is the most critical network metric that needs to be considered. For applications like e-health, intelligent transport system, and structural health monitoring, it is vital to make communication reliable and fast. Therefore, delay in the MAC layer holds a crucial role. Uplink traffic in IoT can be of two types: viz, event-driven and periodic, whereas downlink traffic is mostly triggered, command, and query type. As event-driven traffic is generated when there is an event, this type of traffic remains as rare and valuable. However, if not many changes in the sensing environment of a monitoring application are found, there will be redundant periodic data.

Cognitive: The protocols can reduce interference when multiple networks coexist in a service area, and all of them use the same industrial, scientific, and medical (ISM) band. The use of licensed spectrum is not an affordable solution for a large-scale network due to its high cost. Hence, IoT communication is expected to use a license-free ISM band, which is free of cost. With the large deployment of IoT devices, it is obvious that multiple sub-networks will be deployed in the distance and separately in the same ISM band ([51] and [47]). In such scenarios, frequent interference will degrade the performance of the network. This problem can be solved by using a proper protocol for controlling radios, channels, packet scheduling, etc. There are two types of coexistence mitigation schemes available – *Collaborative* approaches that give coexistence solutions by exchanging information between different co-located radio technologies. In *Non-collaborative* approach, there is no connection between two radios. This technique carries operations like adaptive packet selection, packet scheduling for links, and Adaptive Frequency Hopping (AFH). Cognitive NGI is required in application like intelligent traffic management systems that analyze real-time traffic data, weather conditions, and historical patterns to optimize traffic flow, adapt signal timings, and dynamically reroute vehicles to minimize congestion.

Edge computing: Edge-based computing of making different decisions is essential for low-latency and critical applications [52]. Such computations are helpful for both network and application perspectives. For example, analyzing the sensor data at the edge to take low latency decisions through a new network configuration. Edge

computing is leveraged in smart surveillance systems where video analytics and facial recognition algorithms are deployed at the edge devices to process and analyze video feeds locally, minimizing the need for transmitting large amounts of data to a central server, improving response times, and reducing network congestion.

Caching: It is expected that a considerable number of applications such as vehicular communications and augmented/virtual reality will require caching for data storage at the edge and fog for computations [53]. For example, high-quality graphics/video may need to be stored in a local gateway node in augmented reality applications. In such a case, caching takes a huger responsibility for resource constraint NGI network.

Hyper-connectivity: Hyper-connectivity in IoT is set to drive by 5G communication infrastructure with higher-volume and lower-cost devices. High-Performance computing solutions have been brought to anywhere, directly on their connected devices. Virtualization plays an important role in improving resource efficiency and increasing service reliability, and security [5]. Hyper-connectivity is evident in connected vehicles that utilize Vehicle-to-Vehicle (V2V) and Vehicle-to-Infrastructure (V2I) communication. By exchanging real-time information about traffic conditions, road hazards, and weather, connected vehicles can enhance safety, optimize routes, and enable cooperative driving scenarios.

Heterogeneity and Interoperability: The IoT connects billions of heterogeneous smart devices connected through heterogeneous communications technologies, protocols, and data formats [54]. NGI architecture is anticipated to support these concerning technical and data level heterogeneity. For example, in smart homes, devices from different manufacturers and with diverse communication protocols can seamlessly interact and operate together through interoperability standards like Zigbee or Z-Wave. This allows users to control multiple devices, such as smart thermostats, smart lighting, and home security systems, using a unified interface or voice commands.

3. Comparison of Communication Technologies for NGI

The advances of different wireless technologies help the IoT as a reality. The properties of some of the key enabling technologies along with 802.11ah are compared in Table 4. Furthermore, the application areas of these technologies are shown in Table 5.

BLE: A useful standard for IoT, which has been used widely in many applications, including computing and consumer products [55]. With a significant reduction in power consumption, Bluetooth Low-Energy (BLE) can offer the same range as Bluetooth technology [56]. It works in 2.4 GHz, and the maximum data rate and coverage range are 150m and 1Mbps, respectively.

IEEE 802.15.4/802.15.4g: The PHY and MAC present suitable solution standards for IoT due to its low power, high reliable and secure communication [57]. It operates in the 2.4GHz band and solves requirements or applications that require relatively infrequent data exchanges at low rates over an area within the 100m range.

Upper layer solutions such as 6LoWPAN (IPv6 over Low-power Wireless Personal Area Networks) [58], ZigBee [59], ISA100.11a, WirelessHART [60], SNAP [61], and Thread [62] are based on this standard to fill different requirements of IoT. 6LoWPAN uses compressed IPv6 and provides a solution for IP networking over low power and resource-constraint scenarios. IEEE 802.15.4 and IEEE 802.15.4g have limited communication range, requiring additional network nodes for larger coverage. They operate at low data rates, unsuitable for high-speed applications. Interference from other wireless devices in the same frequency band can degrade network performance. The number of devices supported in a single network is limited, potentially requiring extra infrastructure or network segmentation. Power consumption is a concern, particularly for battery-operated devices, necessitating power management techniques for extended battery life. Both standards are vulnerable to security vulnerabilities and bugs.

IEEE 802.11a/b/g/n/ac: We know these technologies as WiFi-based technologies that are used for Internet connectivity giving lesser importance to constraint communication nature in IoT. These can be used as a backbone for a sensor network in IoT scenarios. However, such deployment again causes more complexity in the network, which is not acceptable in NGI applications.

SigFox/LoRa: Low-Power Wide-Area Network (LPWAN) technologies such as LoRa and SigFox have proved to be suitable standards for IoT to cover a larger distance. These use the ISM band for low-cost solutions. SigFox works in frequency- 900MHz, covering 30-50km in rural environments and 3-10km in urban areas with data rates 10-1000bps [63]. Similarly, LoRa provides data rates range from 0.3 kbps to 50 kbps, covering 2-5km in urban and 15km suburban environments with data rates about 0.3-50 kbps [64].

RFID/NFC: These are omnipresent technologies for IoT, used with a large set of objects and devices around us; they are both instrumental in IoT use cases. Radio-frequency identification (RFID) can connect objects around by tag and reader, further, can able to forward to the Internet [65]. Similarly, Near Field Communication (NFC) is a short-range high-frequency wireless communication technology that enables to exchange of data between devices over about a 10 cm coverage [66]. With very low power consumption and tiny nature, these technologies are the leading enabler for IoT to connect real-world things.

LTE/LTE-A/NB-IoT: Long-Term Evolution (LTE) [67] is designed for high-speed cellular networking based on GSM/UTMS technologies. Improving LTE, LTE- Advanced (LTE-A) [68] is proposed, which provides downlink and uplink multiplexing, larger coverage range, better throughput, and lower delay. LTE-A is proved to be a suitable solution for many cases of M2M communication, but due to the high cost and energy consumption, it can be used for other cases. The excellent battery life and a longer coverage range of Narrow Band-IoT (NB-IoT) [69] show suitability for IoT applications. However, it works on the same LTE band for transmission.

WiFi 6: IEEE 802.11ax wireless communication standard is inherited from 802.11ac, and it is marketed as WiFi 6. A breakneck speed of 9.6 Gbps [70, 38] expecting from WiFi 6 devices which cover relatively less area. IEEE

802.11ah is a better choice for IoT applications covering 1km area with a one hop connectivity, unlike 30-40m in WiFi6. WiFi 6 uses carrier frequency of 20, 40, 80, 80 + 80, 160 MHz. Moreover, enhancing the network performance, different modulation schemes such as BPSK, QPSK, 16-QAM, 64-QAM, 256-QAM, 1024-QAM are used [38].

5G: Fifth Generation (5G) WiFi technology was introduced in 2016, aiming to provide services such as high reliability, network capacity, availability, data speed, energy efficiency, uniformity, and also to provide ultra-low latency [71]. However, this advanced 5G technology still faces challenges while serving Multi-hop communication, high-speed railway communication, multimedia communication, and also in terahertz band channel modelling [72]. IEEE 802.11ah is a better solution for these above mentions issues of 5G.

6G: When the 5G drawbacks are spurring, the next evolution was towards advancing new technology as Sixth Generation (6G) for the applications ranging from autonomous systems to extended reality. The 6G applications such as Multisensory XRConnected, Robotics and Autonomous Systems (CRAS), Wireless Brain-Computer Interactions (BCI), Blockchain, and Distributed Ledger Technologies (DLT) can be marketed significantly [73]. The main features of 6G are the transition from radio to subterahertz (sub-THz) and optical spectra, i.e., visible light communication, Artificial Intelligence and Machine Learning for intelligent computation and network management, and new applications based on new enabling technologies, such as holography [74].

Our objective is to utilize an advanced lightweight MAC protocol for wide area coverage and real-time-based M2M communication. Existing technologies such as LPWAN technologies (e.g., LoRa [64], SigFox [63], NB-IoT [69], etc.) do not have a strong solution [69]. Without sensing the carrier, a device transmits in random access ALOHA, for a lightweight MAC, SigFox and LoRa take this advantage. The NB-IoT architecture operates by utilizing a solitary narrowband RF channel, employing a Time Division Multiple Access (TDMA) mechanism. This design allows for the efficient accommodation of multiple devices on the same channel, enabling synchronized and organized access to the shared communication medium. Through the use of TDMA, NB-IoT optimizes the allocation of time slots, enhancing the overall network efficiency and facilitating reliable communication among numerous devices within its operational range. Multiple hops-based communication technology is enabled in IEEE 802.15.4 [57].

Table 4: Comparison of IoT requirements across various enabling communication technologies

Standard/ Technology	**Year**	**Max. speed**	**Radio frequency band**	**Bandwidth**	**Modulation scheme**	**Range**	**Data rate (max.)**	**Network size**	**Interference**
802.11	1997	2 Mbps	2.4 MHz	20 MHz	DSSS, FHSS	20 (indoor) 100m (outdoor)	-	≈250	Moderate
802.11a	1999	54 Mbps	5 MHz	20 MHz	OFDM (SISO)	35 (indoor) 120m (outdoor)	54Mbps	Upto 64	Low
802.11b	1999	11Mbps	2.4 MHz	20 MHz	DSSS (SISO)	35 (indoor) 140m (outdoor)	11Mbps	Upto 32	Moderate
802.11g	2003	54Mbps	2.4 MHz	20 MHz	OFDM, DSSS (SISO)	38 (indoor) 140m (outdoor)	54Mbps	Upto 32	Moderate
802.11n	2009	150 Mbps	2.4,5 MHz	20, 40 MHz	OFDM (MIMO)	70 (indoor) 250m (outdoor)	450Mbps	Upto 256	Moderate
802.11ac	2013	800 Mbps	5 MHz	20, 40, 80, 160 MHz	OFDM (MU-MIMO)	35m	1.3 Gbps	Upto 64	High
LTE	2004	300 Mbps	LTE band	1.4, 3, 5, 10, 15, 20 MHz,	OFDM	1000m	300 Mbps	Thousands	Moderate
LTE-A	2011	500 Mbps	LTE band	40, 70 MHz	Hybrid OFDM	1000m	3.3 Gbit	Thousands	High
NB-IoT		200 Kbps	LTE band	180 KHz		22km	128 KBit/s	Thousands	Low
802.11ah	2014	78 Mbps	Sub-1GHz	1,2, 4, 8, 16 MHz	OFDM (10 MCSs)	1000m	78Mbps	>8000	Low
802.15.4	2003	250 Kbps	2.4 GHz, 915 MHz, 868.6 MHz	2 MHz (16 Channels)	DSSS	10m	250 Kbps	>100	Low
BLE	2007	1 Mbps	2.4 GHz	2 MHz	GFSK	150m	1-9Mbps	-	Low
NFC	2003	848 Kbps	13.56 MHz	Upto 1MHz	FSK, ASK	20cm	424 kbit/s	2	Low
RFID	1945	640 Kbps	125-135MHz, 13.56MHz, 856-960MHz	Upto 1MHz	ASK, FSK, PSK	100m	100Mbps	>100	Low
SigFox	2009	1000 bps	900MHz	200 KHz	BPSK, GFSK	10 km (urban), 50 km (rural)	600 bps	Millions	Low
LoRa	2015	50 Kbps	868, 915 MHz	220 KHz, 1 MHz	CSS	5 km (urban), 15 km (rural)	27 kbit/s,	Thousands	Low
5G	2016	10Gbps	2.4 GHz, 5GHz	28 GHz, 39GHz	OFDM	1km	20 Gbps	Thousands	High
6G	2020	1 Tbps	140-230 GHz	10THz	BPSK, QPSK	10km	206.25 Gbit/s	-	-
WiFi 6	2021	9.6 Gbps	2.4 GHz, 5GHz	20,40,80,160 MHz	OFDM,BPSK, QPSK, 16-QAM, 64-QAM, 256-QAM, 1024-QAM	30-40m	9.6 Gbps	-	High

Table 5: Standards/Technologies and their application areas.

Standard/ Technology	**Application areas**
802.11	Wi-Fi, Wireless Local Area Networks
802.11a	Wi-Fi, High-speed Wireless Networks
802.11b	Wi-Fi, Low-cost Wireless Networks
802.11g	Wi-Fi, Wireless Networks
802.11n	Wi-Fi, High-performance Wireless Networks
802.11ac	Wi-Fi, Gigabit Wireless Networks
LTE	Mobile Communications, Cellular Networks
LTE-A	Advanced Mobile Communications
NB-IOT	IoT (Internet of Things), Smart Cities, Agriculture, Utilities
802.11AH	Wi-Fi, Sub-1 GHz Wireless Networks
802.15.4	Wireless Sensor Networks, IoT, Industrial Automation
BLE	IoT, Wearables, Beacon Technology
NFC	Mobile Payments, Access Control, Smart Retail
RFID	Asset Tracking, Inventory Management, Supply Chain
SigFox	IoT, Smart Cities, Asset Tracking
LoRa	IoT, Smart Agriculture, Smart Cities
5G	Next-generation Mobile Communications, Smart Cities
6G	Future-generation Mobile Communications
WiFi6	Wi-Fi, High-speed Wireless Networks

However, these relay nodes' duty cycle in 802.15.4 multi-hop is completely busy while routing [75]. Other IoT technologies such as BLE and RFID intend to provide large area communication coverage [76]. These lead to an energy-hungry solution. Upper layer solutions such as 6LoWPAN [58], ZigBee [59], ISA100.11a [77], WirelessHART [60], SNAP [61], and Thread [62] are proposed above this standard to full fill different requirements of IoT. 6LoWPAN uses compressed IPv6 and provides a solution for IP networking over low-power and resource-constraint scenarios. A low-power MAC protocol of IEEE 802.15.4e [78] and Time Synchronized Channel Hopping (TSCH) [75] are the solutions for achieving a low duty cycle and efficient IoT solution, respectively. However, the MAC protocol for IEEE 802.11a/b/g/n/ac provides the facility to transmit more oversized frames that consume high power, resulting in inefficient and unsuitable for IoT. Several innovative ideas of the MAC and PHY layer in IEEE 802.11ah were incorporated, which shows suitability for communications in IoT networks. For the next generation MAC layer designs, we should take into account the traffic heterogeneity concerning load, intervals, and frame size. To be resilient on the scalability concern, we should consider distributed MAC decisions (e.g., at relay nodes or access points). NB-IoT technology has got sufficient attention due to its available channel band (licensed) and frequency band in the market. Based on the recent developments on the MAC layer of IEEE 802.11ah, it not only can associate more than 8000 devices to an AP but also support required channel access among such a huge number of heterogeneous devices due to its dynamic MCSs and RAW configurations support.

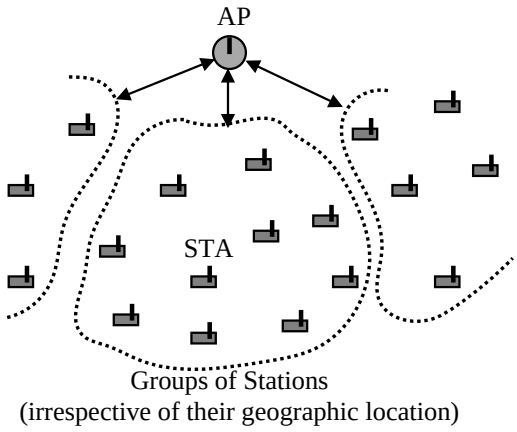


Figure 3: An example of group-based IEEE 802.11ah network of IoT devices

Table 6: Different Modulation and Coding Schemes (MCSs) and their data-rate supported by IEEE 802.11ah

MCS	Modulation	Coding rate	1 MHz (Mbps)	2 MHz (Mbps)	4 MHz (Mbps)	8 MHz (Mbps)	16 MHz (Mbps)
MCS0	BPSK	1/2	0.3	0.65	1.35	2.925	5.85
MCS1	QPSK	1/2	0.6	1.30	2.70	5.850	11.70
MCS2	QPSK	3/4	0.9	1.95	4.05	8.775	17.55
MCS3	16QAM	1/2	1.2	2.60	5.40	11.700	23.40
MCS4	16QAM	3/4	1.8	3.90	8.10	17.550	35.10
MCS5	64QAM	2/3	2.4	5.20	10.80	23.400	46.80
MCS6	64QAM	3/4	2.7	5.85	12.15	26.325	52.65
MCS7	64QAM	5/6	3.0	6.50	13.50	29.250	58.50
MCS8	256QAM	3/4	3.6	7.80	16.20	35.100	70.20
MCS9	256QAM	5/6	4.0	-	18.00	39.000	78.00
MCS10	BPSK	1/2	0.15	-	-	-	-

These features of 802.11ah have proven as a solution to high collisions of packets and lows scalability problems in IoT networks ([25, 79]). IEEE 802.11ah encloses a blended approach of MAC and multiple configurations of PHY solutions in a RAW frame, where contention, reservation, MCS, channelization, etc., schemes are used for slot allotment in order to perform successful transmission. In the channel access mechanism of IEEE 802.11ah, a set of stations are grouped together. Thus, multiple groups are formed to minimize collisions. As shown in Fig. 3, the AP creates the groups, and only one of them is allowed to access the channel in a particular RAW frame. The below section discusses the features of IEEE 802.11ah for Next-Generation IoT.

Table 7: Expected PHY Layer features for NGI

PHY features	NGI Features	Detail
Multiple MCSs	Heterogeneous load, Dynamic bandwidth, Scalability, Efficiency	Supports MCS0 to MCS10 for data-rates 150Kbps to 78Mbps
Multiple Channels	Coexistence, Scalability, collisions	MCS use multiple channels and bandwidth
Sectorization and Beamforming	Scalability, efficiency, QoS	Sectorized antenna support for MIMO and beamforming
Sub-1 GHz Channel band	Energy efficiency. congestion, coverage	Sub-1 GHz can cover upto 1 Km in a hop
FFT	Coexistence, Interference	FFT is used to generate OFDM symbol
Multi User MIMO	Scalability, reliability	MIMO with sector antennas for AP and relay, enhances network capacity

4. Enabling feature of 802.11ah for Next Generation Internet of Things

IEEE 802.11ah-based wireless communication (license free) technologies solution seem to be the best option for NGI [3]. It supports large coverage areas, energy efficiency, scalability, heterogeneity, controlling huge contention, and minimize collisions. Moreover, IEEE 802.11ah minimizes the protocol overhead for short frames with adaptive bandwidth to the changing demands and reduces interference when multiple networks and relays. The enabling PHY and MAC layer features are discussed below:

4.1. PHY Layer Enhancements

The IEEE 802.11ah amendment aims to provide wireless LANs (WLANs) with the ability to both manage a large number of heterogeneous stations within a single Basic Service Set (BSS) and minimize the energy consumption of the sensor-type battery-powered stations. The Task Group of 802.11ah (TGah) came up with the concept of using multiple MCSs in different conditions to serve heterogeneous traffic requisite of large-scale IoT applications. Table 6 shows the MCSs and output data rates. For example, a 16MHz channel band with MCS9 yields a data rate of maximum 78Mbps. PHY layer features are discussed against NGI requirements in Table 7.

The channelization in 802.11ah inherits the PHY layer of 802.11ac including the concept of Null Data Packet (NDP) [80]. The wireless spectra available in each country are the key factors in determining the channelization for that particular country. Fig. 4 gives the available license-exempt channels in sub-1 GHz spectrum for different countries, for example, 902-928 MHz (US), 863-868 MHz (Europe), 917.5-923.5 MHz (South Korea), 755-757 MHz (China), 865-867 MHz (India), 916.5-927.5 MHz (Japan), and two non-contiguous frequency bands are utilized by Singapore i.e., 866–869 and 920–925 MHz bands [80, 81, 82]. The channel bandwidth in 802.11ah includes 1, 2, 4, 8, and 16 MHz, whereas 1 and 2 MHz are widely available in many countries and more suitable for low rate traffic scenarios. The channelization in 802.11ah allows for concurrent communication on multiple channels, enabling higher network capacity and improved scalability for IoT deployments. It also helps mitigate interference from neighboring

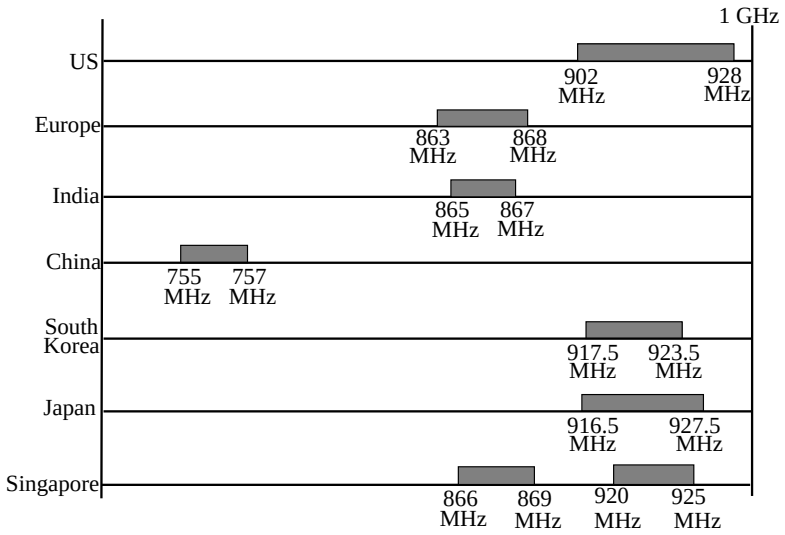


Figure 4: Availability of country-wise licence-exempt bands in sub-1GHz

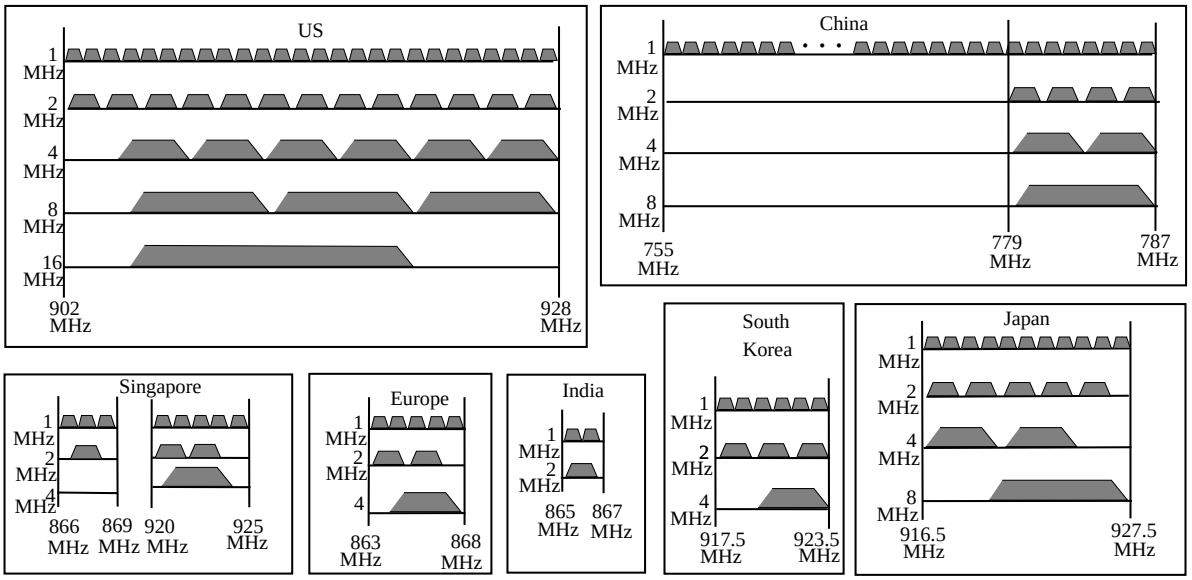


Figure 5: Country wise different channels from usable Sub-1 GHz bands

networks operating in the same frequency band. The use of narrower channels (e.g., 1 MHz) allows for better spectral efficiency and increased range, making it suitable for long-range IoT applications. However, the trade-off is lower data rates compared to wider channel widths used in other Wi-Fi standards.

Fig. 5 depicts the possible channel bands in different countries [80, 81, 82]. The US's IEEE 802.11ah channelization has received significant attention due to its allowance of up to 16MHz bandwidth between 902MHz and 928MHz [80]. It includes channel bandwidths of 1MHz, 2MHz, 4MHz, 8MHz, and 16MHz. However, IEEE 802.11ah has adopted the minimum channel bandwidths of 1 MHz and 2 MHz, and it is necessary for the IEEE 802.11ah standardization group to support these bandwidths for the US and other countries being considered. In Europe, there are 5 usable channels with 1MHz bandwidth, 2 channels with 2MHz bandwidth, and one channel with 4MHz bandwidth. For channelization between 865-867MHz, India adopted 2 channels with 1MHz bandwidth and one channel with 2MHz bandwidth. China allows a maximum permitted transmitting power of 5mW for frequencies from 755MHz to 779MHz, and 10mW for those from 779MHz to 787MHz within the range of 755-787Mhz. In the 10mW capability range, China permits 4 channels with 2MHz bandwidth, 2 channels with 4MHz bandwidth, and one channel with 8MHz bandwidth. South Korea has 6 usable channels with 1MHz bandwidth, 3 channels with 2MHz bandwidth, and one channel with 4MHz bandwidth in the range of 917.5-923.5MHz. In the range of 916.5-927.5MHz, Japan can use 11 channels with 1MHz bandwidth, 5 channels with 2MHz bandwidth, 2 channels with 4MHz bandwidth, and one channel with 8MHz bandwidth. Singapore has two different frequency ranges, and combining both, a total of 8 channels with 1MHz bandwidth, 3 channels with 2MHz bandwidth, and one channel with 4MHz bandwidth are allowed to be used.

Using sub-1GHz (license-exempt) channel bandwidth has advantages such as reliability and longer battery life in addition to covering a more considerable distance. The spectrum has sounding propagation and penetration for large coverage areas in one-hop reach. Considering the current deployment of other license-exempt channels, sub-1 GHz is expected to be a solution for lesser congested frequency band, high sensitivity, and link margin available diversity in terms of frequency, time, space. Moreover, it is suitable for long battery life and short data transmissions. Therefore, the sub-1 GHz frequency band is considered one of the most appropriate communication mediums for the NGI perspective. To increase the wireless coverage range, MIMO modifications are proposed to exploit precoding features [83]. The proposed modifications include the use of modified Preparata codes, including the *Grassmannian* and *Kerdock* manifold. Additionally, beamforming in 802.11ah is the key for signal transmission techniques to mitigate wireless interference in long-range outdoor scenarios. The use of MIMO provides insights into the transmission performance of concurrent flows when beamforming is used.

Fig. 6 shows an architecture of 802.11ah, which uses sectorized beamforming for APs and relays. Multiple sector antennas can be sued at relays to improve the network capacity, which is essential for NGI applications.

4.2. MAC Layer Enhancement

The 802.11ah standard primarily inherits the basic features from the 802.11ac, which is further inherited from the traditional 802.11. IEEE 802.11ah standard is powered by a set of new and novel concepts. After the first WiFi standard, IEEE 802.11, was proposed in 1997, a series of advancements have been made. Standards such as 802.11n,

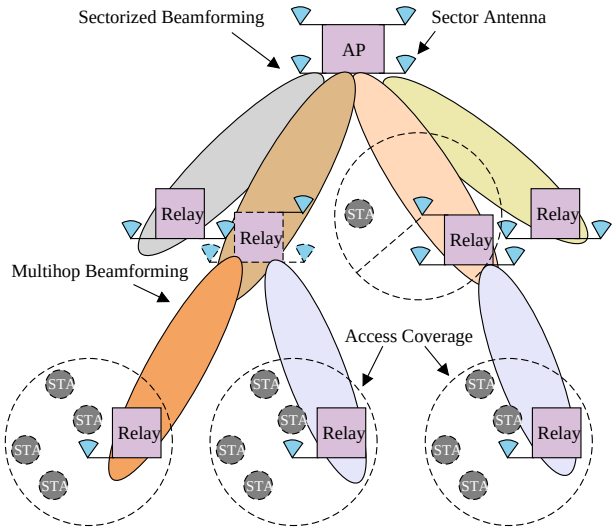


Figure 6: An example of sectorized beamforming in relays and AP for IEEE 802.11ah networks

Table 8: Expected MAC layer features for NGI

MAC features	**NGI Features**	**Description**
Hierarchical AID	Scalability	Mitigating collision during link set-up
RAW	Scalability, Efficiency	Reduces collsioon domain during contention
AID and grouping	Scalability, Energy efficiency, Organization of stations	QoS, Management
TIM Segmentation	Energy efficiency	Stations waking up less for receiving beacons
TWT	Energy efficiency	Station negotiating with AP about wake-up time
BSS Color	Scalability	Reducing interference among OBSSs
Short MAC header	Reducing overhead	Using 2-byte address, and containing less subfields
Response Indication Deferral (RID)	Carrier sensing	Carrier sensing when Short MAC header are used
Relay	Range extension	Two-hop link between a station and the AP

802.11ac, and WiFi-6 mainly tried to improve network throughput. In IEEE 802.11ah, the new features such as RAW, AID, and TIM and Segmentation mechanisms, help to connect thousands of stations in a single network which is very important in NGI. The MAC layer features are presented against the NGI features as shown in Table 8.

Due to the limitations in AID field size, IEEE 802.11 standards had a maximum capacity of associating up to 2007 stations to an access point before its advancements to WiFi-HaLow. With a hierarchical structure, 13-bit AID is used in 802.11ah for each associated station. The structure is hierarchically divided as Pages:Blocks:Sub-blocks:Stations. A TIM block contains a set of stations with AIDs of a particular range. Thus, in 802.11ah, a single access point can accommodate up to $2^{13} - 1$=8191 stations. The hierarchical AID organization is shown in Fig. 7.

IEEE 802.11ah proposes new power-saving schemes to reduce the energy consumption of sensor devices. These

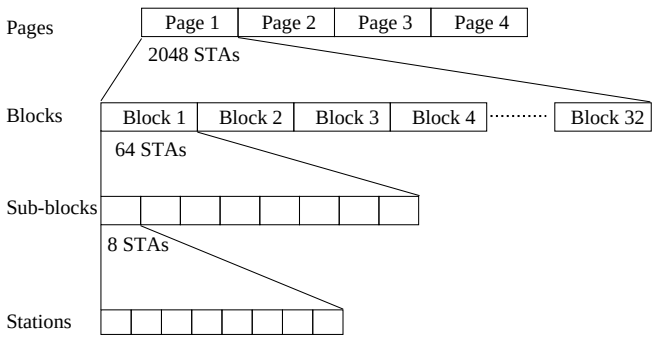


Figure 7: Hierarchical AID organization of IEEE 802.11ah

schemes use the types of the station while scheduling their radio on/off duration. This mechanism reduces the waiting time for channel access and increases the sleep duration of stations. Here, the channel access time is hierarchically converted into slots as: Time → Pages → Delivery TIM (DTIM) periods → TIM periods → Slots. When an access point broadcasts the DTIM and TIM beacon to the associated stations, the TIM and DTIM period is counted. The access point sends a DTIM beacon informing about the pending data to the TIM groups within. Then, the TIM beacon carries in the information of which station has pending data at access point. If there is no pending data at access point; it will keep itself in the power-saving state by following two conditions – DTIM beacon notifies the station of TIM group that there is no pending data at access point, and station knows about pending data at access point but is not explicitly aware of the TIM beacon.

As compared to the traditional power saving scheme of 802.11, the advantages in 802.11ah reduce – energy consumption for stations not having data at AP, the traffic indication map size in TIM beacon, and customized size of DTIM for better energy efficiency. IEEE 802.11ah technology is likely to be adopted for its strong grip on energy-saving mechanisms. A station need not be awake all the time; instead, it can communicate with the access point, and using the Target Wait Time (TWT) mechanism, it finds its next wake-up time. For TIM group paging in a huge IoT environment, the stations may unnecessarily wake up.

Using the existing channel access mechanism, communication among thousands of stations increases the chance of a collision to a large extent. To this problem, 802.11ah promises to increase throughput and energy consumption using a RAW mechanism. It divides the stations into different RAW groups that spread channel access over a long time. The RAW is further divided into slots; a station can only run its Transmission/Reception (Tx/Rx) mechanism in the dedicated slot. The stations belonging to a RAW group are allowed for contention within the assigned slot duration.

The channel access mechanism in 802.11ah is shown in Fig. 8. The channel access frame is divided into RAWs,

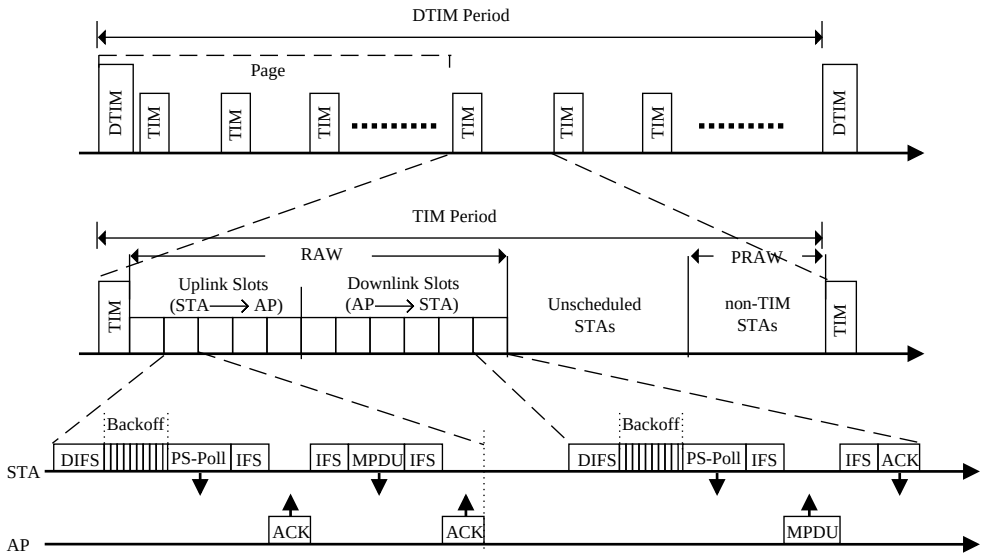


Figure 8: Hierarchical TIM and RAW based channel access mechanism in IEEE 802.11ah

and a group of stations is allowed from transmission within a RAW. At the start of RAW, AP sends broadcast beacons carrying RPS (write full form of RPS). Stations (STAs) can know about the availability of channels and AP to start a transmission. Further, a RAW frame can be further divided as per the type of STA and traffic patterns. For example, uplink/downlink, periodic and non-periodic, etc. A STA wants to start a transmission sends a PS-Poll message after completing the default backoff process. Once a PS-Poll is successful, the STA can send or receive a frame. At the beginning of RAW, AP broadcasts a synchronization (SYNC) frame. The STAs can detect the availability of the channel and can initiate its transmission. Hence, the STAs do not need to wait for an allocated time frame called *ProbeDelay* timeout [84]. Furthermore, we discussed the key features of IEEE 802.11ah below:

EDCA: Enhanced Distributed Channel Access (EDCA) is adopted by 802.11ah for service category based access [85]. It provides traffic-aware channel access employing different inter-frame spaces for each category. A flexible time duration named Arbitration Inter-Frame Space (AIFS) is used before the backoff period. Accordingly, the contention period is not the same for different types of traffic classes.

Periodic RAW (PRAW): In IoT, a huge number of STAs may be deployed for monitoring purposes. The STAs mostly follow periodic traffic patterns. To reduce overhead for such traffic, 802.11ah proposes a PRAW as a part of the existing RAW frame. If the periodicity is known, STAs can be scheduled in the PRAW phase without requiring multiple times of contentions. These reduce overall access time, which further increases the efficiency of the network. The RPS may indicate the PRAW allocation. The allocated indication is further broadcasted by AP

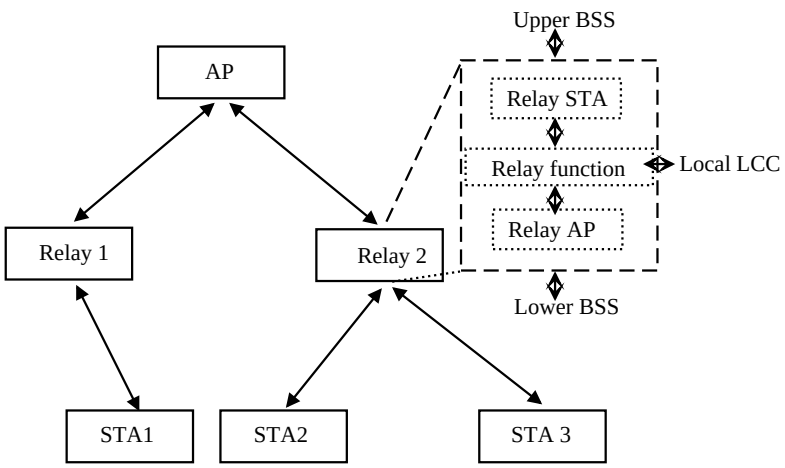


Figure 9: IEEE 802.11ah-based multi-hop networking architecture with relay nodes

such that all other PRAW STAs can know about the scheduled details.

Cross-slot boundary: If an STA is transmitting or counting down in backoff over EDCA, there is an option for AP to allow the transmission over the cross-slot boundary (CSB). The CSB indication is sent using RPS with 1bit filed. If the CSB field is "0", an STA can cross the slot boundary; otherwise (i.e., CSB field "1"), there is not enough space in the RAW, which is called the non-cross boundary (non-CSB) case.

4.2.1. Relay Node Support

IEEE 802.11ah standard provides RAP support which works as a relay between AP and STAs. It extends connectivity for an STA located outside the coverage of AP. IEEE 802.11ah implements RAP consisting of an AP and an STA module in it [86]. A *Relay function* is responsible for connecting AP entity with STA entity. Figure 9 shows network architecture with relay nodes. Frames coming from AP are received by Relay STA and forwarded to STA 3 via Relay AP.

IEEE 802.11ah is a dominant savior for seamless communication in large-scale IoT environments. The channel bandwidths – 1 MHz and 2 MHz and coding schemes–MCS0 and MCS10 are widely used in NGI for short burst data-rates and long-range communication ([21] and [87]). A list of organizations such [88], [89], and [90], are trying to implement the features in their 802.11ah-based products. We discussed the prominent features of IEEE 802.11ah with reference to the existing research. Moreover, a list of research works can be found that discuss the idea of MAC and PHY layer concepts in detail. Fig. 10 highlights the key enabling features of NGI architecture.

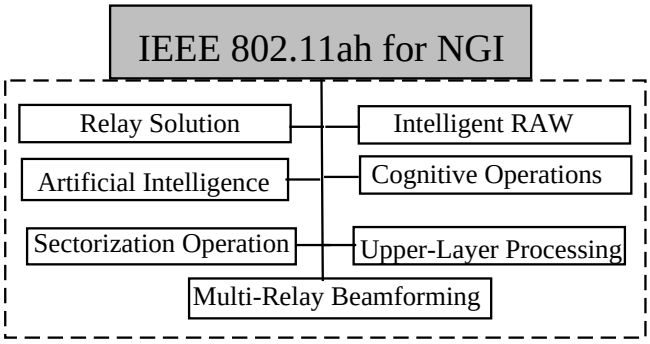


Figure 10: Key enabler for NGI alongside IEEE 802.11ah networks

5. IEEE 802.11ah-Based Next Generation IoT Architecture

IoT is dedicated to serving more intelligent services while using heterogeneous technologies fitted to sense, collect, process, analyze and manage data [91]. The research community is immensely eager to bring next-generation WiFi technology to the table for its stupendous features like support of heterogeneity along with enabling AI and edge computing plus distributed security befitting for gigantic networks. To enhance the automated data processing power, the upcoming technology for IoT is to adopt machine learning and neural network technology of AI. Internet of Things is considered as the key data provider for artificial intelligence. In the view of new IoT technology development aiming fast, robust, low cost, and meaningful, it is to use edge computing for smart and effective computation in real-time scenarios along with felicitating low latency service. Edge computing is a solution to the fastest data analysis for sensed data by IoT devices [92]. IEEE 802.11ah or WiFi HaLow is a novel IoT-enabled WiFi technology to tackle the drawbacks facing by IoT while providing connectivity among a large number of devices with limited energy placed in a wide geographical area. IoT communication problems can be addressed by handling complexity and coexistence and making the network easier to manage using 8022.11ah based technology in an IoT environment.

For advanced coexistence management along with flexible infrastructure in application scenarios such as industrial IoT and smart grid architecture, IEEE 802.11ah plays a key role [93, 94]. It is not wise to use solutions provided by IEEE 802.15.4 for coexistence management problems for assigned channels of the WiFi standard i,e. 2.4GHz [22].

We formally define IEEE 802.11ah-based NGI architecture as a dynamic infrastructure capable of reconfiguring the network and application requirements through programming remotely. It must provide the facility to connect more number of devices and to cover a vast geographic area. Moreover, reluctant to serve varieties of devices with different technology and protocols associated. Managing the network by having the network overview of the architecture is a key service of NGI. In NGI, we also expect to have the facilities of grouping/regrouping, coexistence and the optimized energy consumption parameter. An overview of the proposed NGI architecture with 802.11ah

can be seen in Fig. 11. The proposed network solution connects remote sensor and actuator nodes using different components such as Relays, MCSs, AID, etc., as supported by IEEE 802.11a. The components facilitate by our proposed IEEE 802.11 based NGI architecture are shown in Fig. 12 and detailed as follows:

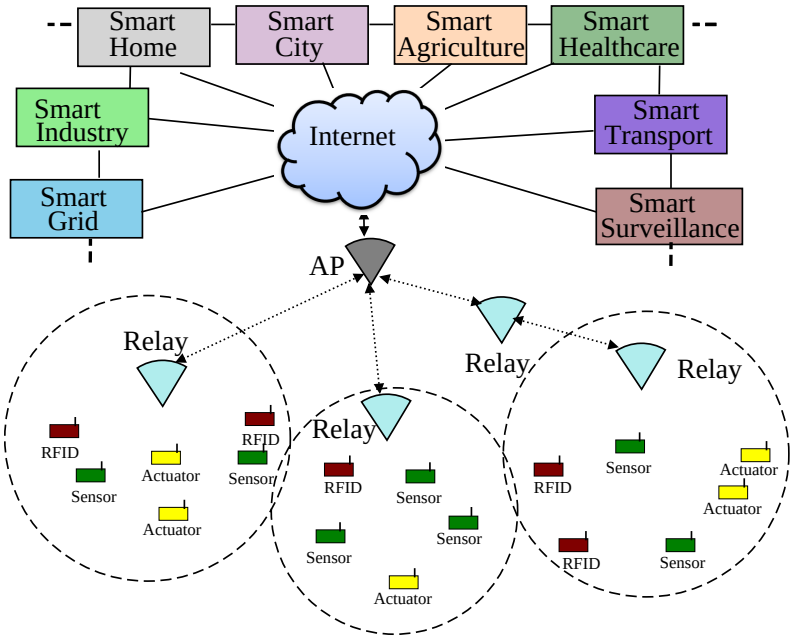


Figure 11: Proposed IEEE 802.11ah-based NGI Architecture

5.1. Edge Computing

With the rapid growth of technology, we expect the devices to process and compute data as well as take appropriate decisions in milliseconds. In such a case, we adopt the technology as edge computing by providing the computation at the network edge devices, which are nearest to the end-user. IEEE 802.11ah with long-distance links and multi-hop communication may increase end-to-end latency. Therefore, provisioning low latency communication for critical IoT applications such as industry, healthcare, etc., is a challenge. As such, the considered infrastructure with edge computing could provide services with minimal latency, a lot of time-sensitive computation services are benefited. Load balancing of network traffic at the distributed network structure, the edge devices while ensuring low latency service, reduces the burden on the centralized cloud server [95]. Edge computing in IEEE 802.11ah also benefits a network by handling congestion as each edge device takes care of the traffic generated from the nearby devices. In this paradigm, while the computation is at the nearest device from the sensing object, it ensures privacy of data as the data is not needed to send to a remote cloud over the Internet, and for the same reason, the system increases

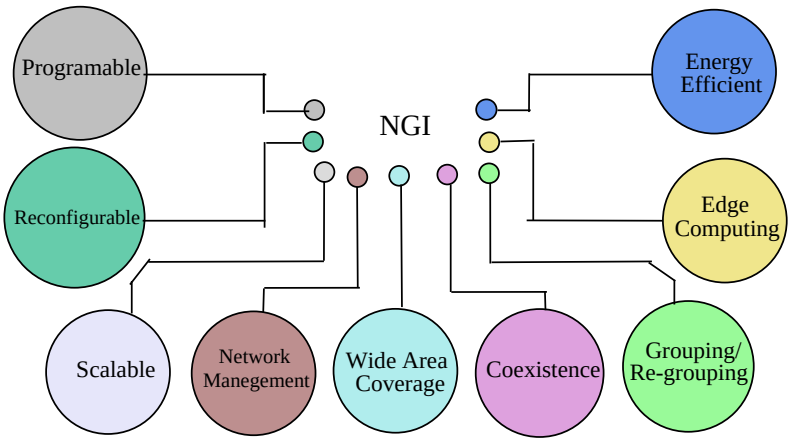


Figure 12: Different components of NGI

the battery lifetime of the devices associated with each computing node [96]. Edge computing also has a great significance for minimizing resource utilization in resource constraint IoT networks such as energy and bandwidth. In instances such as edge computing in healthcare can be an extremely crucial mechanism to save lives by taking decisions in milliseconds while computation is done in local devices rather than at a remote cloud. Edge computing is adopted widely in other time-critical network services such as industrial IoT networks, vehicular networks, and smart-city.

5.2. Re-configurability

Recent technological developments have modernized the old healthcare, agriculture, and most industrial sector from autonomous to automation [97]. It is expensive to reconfigure a computing device once it is deployed in a wide geographical area to change types of services. The reconfigurability in a network smartly configures to provide various existing or new task types dynamically. Considering resource constrain on IoT devices, it is vital to optimizing resource utilization of the devices, which can be achieved by reconfiguration mechanism[98]. IEEE 802.11ah has proposed many flexible network parameters such as MCS, RAW, TIM, TAW, etc., for supporting the dynamic requirements of a network. For example, MCS can be changed dynamically for better channel usage and data rate. Similarly, a new RAW configuration is created by the AP based on the traffic demands in the network. Moreover, the end devices can be reprogrammed for specific data delivery and sleep schedule configuration. Instead of declaring the variables locally, in the proposed architecture, the variables are arguments for hosting such parameters remotely for efficient utilization of channels and increased data rate.

5.3. Network Management

IEEE 802.11ah provides a multiple-channel access facility through which users may select the appropriate channel for smooth usage. Availing the multi-hop feature in 802.11ah based IoT network, we may expand our coverage and manage the extended large network dynamically from a remote place. Managing the IoT network using different MCS's to control the traffic congestion is a significant attribute of this technology [99]. IEEE 802.11ah based NGI is to have a global view of the network, and it should be able to configure/reconfigure the network requirements. Moreover, 802.11ah is considered for connecting a large number of devices placed at the sensing/actuation field and backbone networks. Therefore, complexity and heterogeneity in terms of networks, protocols, and packet formats have been automatically reduced. Managing such a network becomes easy with its growing size.

5.4. Scalability

The proposed architecture is anticipated to work with a large number of devices connected to the network, and as users need it, it should be able to associate more number of devices. Scalability is an essential feature of the NGI system that has the capability to handle network services. IEEE 802.11ah technology has a novel technique as a RAW-based channel access mechanism to achieve scalable and reliable communication among a huge number of heterogeneous devices. Along with a multi-hop facility, a hierarchical network topology is constructed with a huge number of devices to improve scalability. Millions of devices to be associated with an NGI architecture should take the scalability feature into account. In 802.11ah, a fast device association mechanism is developed, which minimizes collisions to raise scalability. However, ensuring scalability is a challenging task for an IoT environment as services like identity management of new devices is a concern of network security and protocol implementation. Also, maintaining the privacy for adding new devices is a significant research concern [100]. As 802.11ah ensures scalability, using which we may achieve another key advancement for NGI.

5.5. Coverage

In the IoT network paradigm, the stations are saturated over a wide area. Hence, the network should provide services to that area considering all the devices placed randomly. To address the challenges of covering a large area, existing WiFi technology could serve only up to a few hundred meters. With the recognition of 802.11ah with a multi-hop network facility and sub-1GHz channel band, it can cover at most eight thousand devices placed within a range of 1km using a single hop, which can be more with multi-hop scenarios. However, 802.11ah is limited to a maximum of two hops stations. Indulging this long-distance facility of 802.11ah to the NGI architecture, we can provide low-cost, effective services in smart-city like applications.

5.6. Coexistence

Features of 802.11ah, such as more extended coverage, Orthogonal frequency-division multiplexing (OFDM), Fast Fourier Transform (FTT), and relay placement, help the network significantly reduce coexistence. As it is new and uses sub-1 GHz, fewer other communication technologies use the same frequency bands in the currently deployed area. Intelligent frequency modulation techniques such as OFDM, FTT, etc., helps to reduce such interference significantly.

5.7. Grouping/Re-grouping

Using a device association mechanism, namely AID in 8021.11ah, individual devices in that network have a four-level structure. Those four levels are page, block, sub-block, and stations. Groups are formed based on the similarities they are holding in respect to the mentioned four levels. Also, to be noted that the association is made in such a way that the devices can be de-link from one group and join with other nearby groups. The proposed RAW and DTIM mechanism also considers the grouping of stations for improving scalability and reliability.

With the recognition of the 802.11ah, we strengthen the NGI architecture towards a more flexible network architecture.

5.8. Energy Efficiency

In IoT, devices are densely deployed in a random manner, and these devices are mostly battery-operated. When the devices with limited energy are placed once, it is more unlikely to reuse once the energy is totally consumed. This crucial challenge of minimizing energy consummation in order to extend the lifetime of the devices, researchers put a lot of efforts[101, 102, 103]. IEEE 802.11ah uses two novel techniques, TIM and DTIM, for access points to communicate the group information and for the associated devices for their next transaction. In the beginning, TIM was introduced to let the devices of a particular TIM group communicate with an access point on that time interval. TIM group devices respond to the beacon if they received the TIM signal very recently; otherwise, the devices go on sleep mood to save energy [24, 104, 84, 105, 106]. Sending TIM signal to the devices very frequently may be a cause of energy conception; to solve this, DTIM was introduced to send periodically. Moreover, IEEE 802.11ah advances its own energy efficiency by introducing TWT mechanism. When some devices communicate very rarely at a definite time, in that case, TWT is used so that the device wakes up at the targeted time to send data to the access point; otherwise, the device remains in sleep mode.

6. IEEE 802.11ah-based IoT stack

The NGI encompasses a broad spectrum for the use over constraints and competent (non-constrained) devices such as sensor/actuator and edge/fog, respectively. The end devices have limited energy resources regarding processing

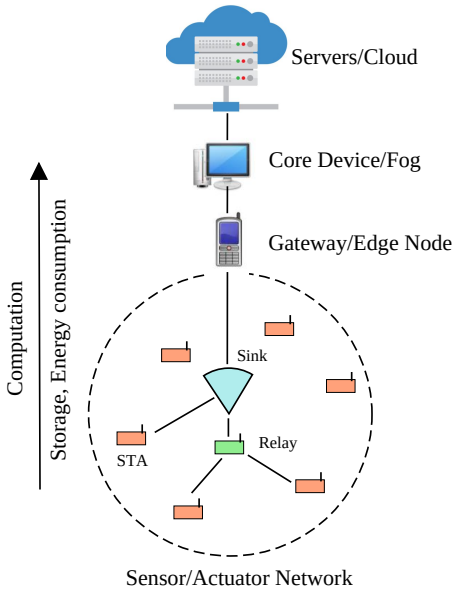


Figure 13: Deployment of heterogeneous devices and their position in IoT

and communication compared to the devices towards Cloud. Fig. 13 shows the position of devices as per their computing, energy consumption, and storage capabilities. Communication requires a huge energy power for local processing and transmission frames. Also, protocols to be used in a different layer of IoT protocol stack should also be concerned about the resource constraint nature of devices. Link-layer has a significant impact because it is responsible for the nuances of the physical transmission technology, framing, media access control (MAC), and re-transmissions [107]. Therefore, with the help of the new link-layer (PHY and MAC) support of IEEE 802.11ah, a 6LoWPAN-based upper layer solution can be a suitable option for Next Generation Internet of Things communication. 6LoWPAN is an IPv6-based low-power WPAN that runs over the devices that supports IEEE 802.15.4. It allows interoperability between the IPv6 domain using an adaptation mechanism adopted by it. In the case of IEEE 802.11ah, it can not provide interoperability and other lightweight upper layer activities such as mesh routing, end-to-end security, and application processing. The current works on 802.11ah define only a star type topology at link-layer connectivity. Here, any communication is made through the AP node. The popular mesh topology is not supported in this standard. Also, communication between the AP nodes is not supported directly. IPv4 and IPv6 are compatible with 802.11ah via the logical link control (LLC) [108]. Fig. 14 present the possible IoT for supporting different lightweight NGI protocols and applications.

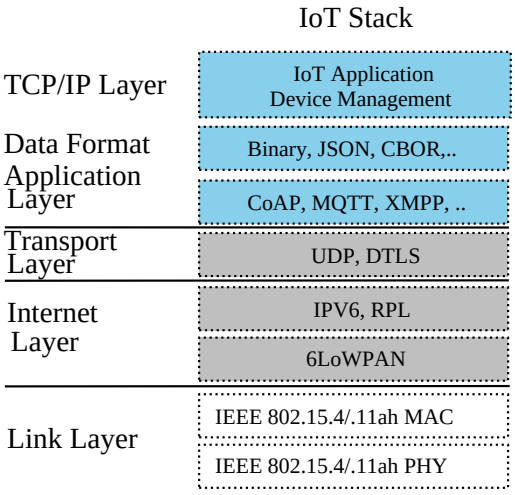


Figure 14: IEEE 802.11ah-based NGI stack for supporting lightweight protocols

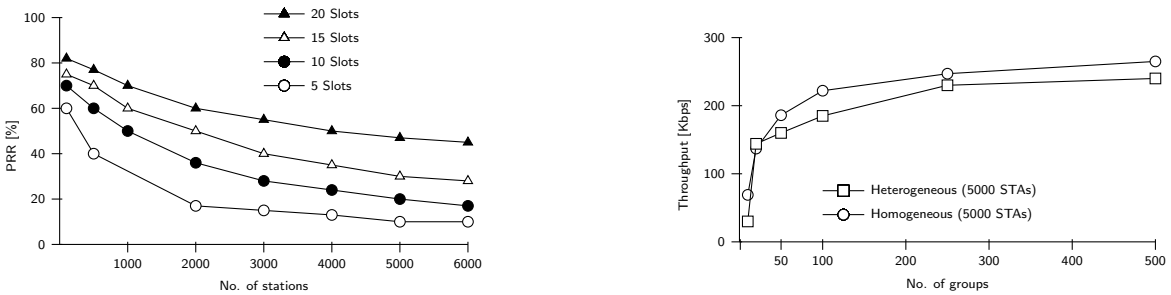
7. Performance Evaluation

We evaluated the performance of the proposed network architecture through simulations conducted using the NS-3 simulator [109]. For our experimental network, we considered a maximum of 6000 stations with up to 3 hops and a maximum of 15 groups. Each experiment was conducted within a service area spanning $2000 \times 2000m^2$ and lasted for a simulation time of 5 minutes. As shown in the network architecture (refer to Figure 6), stations are distributed within 2km square regions, utilizing relays to extend connections beyond 1km. The distribution of nodes is random, following macro propagation model for communication. Given that the proposed network covers 2km through relay nodes, we anticipate that the communication setup is well-suited for rural connectivity. In the NS-3 simulation setup, various network parameters were taken into account, including bandwidth, data rate, payload size, and distribution type, as well as the path loss model. The parameters and their default values are listed in Table 9. The key performance metrics of the analysis are as follows:

- **Delay:** It is the time taken by a frame to reach the access point from a station and vice-versa.
- **Coverage:** We measure the successful packet delivery with increasing hop distance and number of hops.
- **Throughput:** It is calculated by monitoring the number of packets received at the destination over time.
- **Packet loss:** It is the ratio of the number of frames received at the destination over the total of frames sent.
- **Control-loop latency:** To check the performance of different automation scenarios of IoT, we check the round trip time (RTT) of the frame considering edge-based decisions at the relays and AP.

Table 9: Parameters used in simulation and analysis.

Parameters	**Value**
Bandwidth	$2MHz$ (MCS0)
Data rate (δ)	650Kbps
Payload size (l) /Traffic type/ Traffic rate	256Bytes/UDP/ ≈2Kbps*
CWmin/CWmax, σ/σ_{max}	15/1023, 0.5/0.7 Sec.
Backoff slot /SIFS time /DIFS time	52 /160 /SIFS+2*slot time (μs)
Distribution / Path loss model	Random /Macro [110]
δ_s/t_s/Coding rate BPSK-MCS0 (γ)	26 bits/40 μs /0.5
t_{phy} / MAC (m_h) header	6 * symbol duration (μs) /14Bytes
Queue/CQ size	100 /10
Hops/Group/RAW size/No. of STAs (Max.)	1/2-10 /15/6000
$p_{tx}/p_{rx}/p_{id}/p_{sl}$	255 /135/135 /1.5mW [111]
Simulator /Simulation area /Time	NS-3** /2000×2000 m^2 /5 Minutes.
*Traffic Interval= 1 Sec.	**https://www.nsnam.org/



(a) (b)

Figure 15: (a) Packet Received Ratio (PRR) for different RAW sizes with increasing number of STAs, (b) Throughput for different group sizes in homogeneous and heterogeneous traffic conditions

A bigger RAW size performs superior with a higher number of associated stations when we transmit frames for a specific period. We have analyzed the performance of RAW size with the increasing number of STAs. As shown in Fig. 15a, PRR (%) decreases with a growing number of STAs. With the same RAW size, when more number of STAs try for contention, collisions is higher. However, PRR (%) is higher for larger RAW sizes because the chances of getting a slot are better in this case. Also, we have analyzed the throughput performance over heterogeneous traffic scenarios. For heterogeneity, traffic is generated by sending packets in 0.1, 0.5, 1, 5 Seconds intervals. Fig. 15b shows the throughput achieved in traditional RAW with an increasing number of groups over homogeneous and

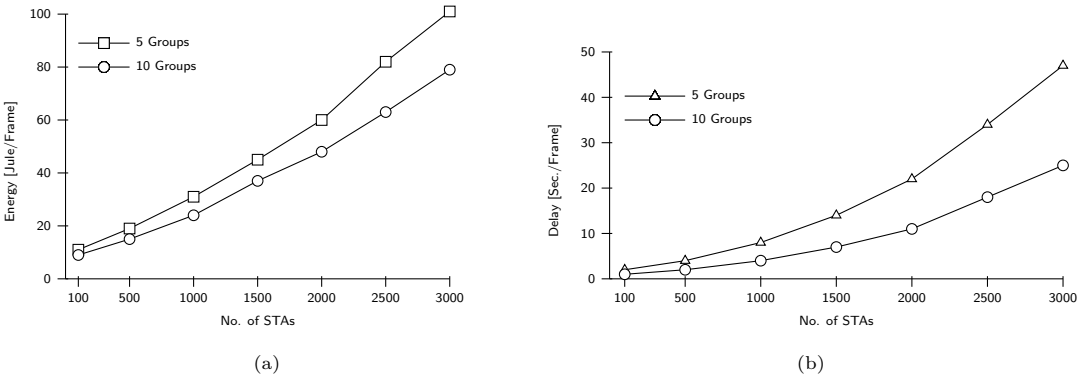


Figure 16: (a) Energy consumption, (b) Delay considering considering TIM STAs and RAW size as 15 slots

heterogeneous network scenarios. For 5000 number of STAs, a higher number of groups reduce contention within a group. However, when traffic is transmitted at different intervals, throughput performance deteriorates. As an STA may or may not have traffic to transmit in a group, for fixed-size RAW, throughput performance degrades. A research lacuna exists to predict proper RAW size under heterogeneous network conditions. Different from the homogeneous scenario, heterogeneous networks are defined by not only traffic intervals but also parameters like sleep time and QoS requirements which brings an enormous design space. As shown in Fig. 16a, when there are lesser groups, a large number of nodes simultaneously try for contention and hence collisions occur. Due to the re-transmissions, the average per-packet energy consumption increases drastically with the increasing number of STAs. However, when there is more number of groups, the chances of collisions are less. A performance analysis is carried out to see the impact of hidden nodes. The STAs are randomly grouped irrespective of their geographic locations, which are spread over 1Km×1Km and 2Km×2Km. As an STA can cover up to 1Km of distance, the number of hidden nodes for the larger coverage area will be higher. As shown in Fig. 17a, throughput for TIM STA increases with the increasing number of groups. However, in the case of more extensive coverage, the performance is lesser. This is due to the fact that with increasing group size and coverage range, the number of hidden nodes is reduced and increased, respectively. Again, with non-TIM and unscheduled STAs, throughput decreases drastically with an increasing number of nodes, especially in the case of 2Km×2Km area (can be seen in Fig. 17b). NB-IoT architecture uses a single narrowband RF channel and employs Time Division Multiple Access (TDMA) to accommodate multiple devices on the same channel. In scenarios with a higher number of nodes within a single group, the impact on the number of hidden nodes is generally minimal unless new nodes are introduced into distant locations. However, there can be an increase in hidden nodes, leading to increased time consumption, primarily due to the delay in receiving RTS/CTS frames. The number of hidden nodes diminishes with increasing group size, but it rises with extended coverage, affecting overall performance. Additionally, performance can be significantly improved in IEEE 802.11ah

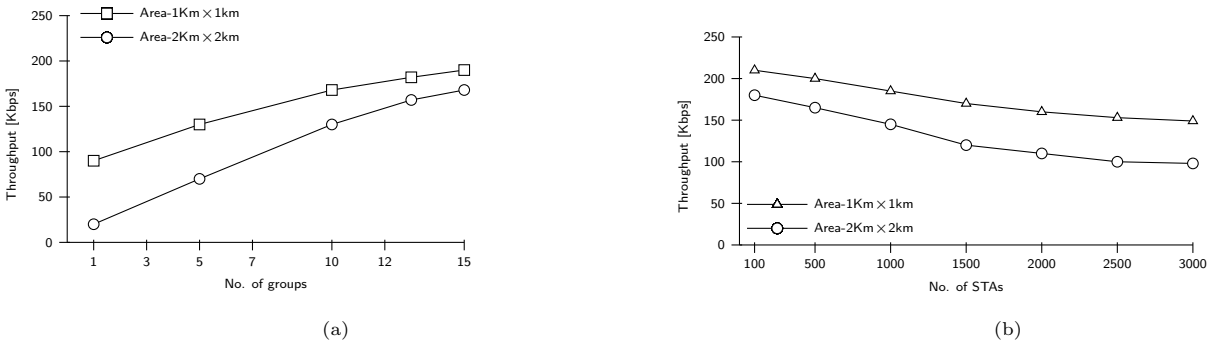
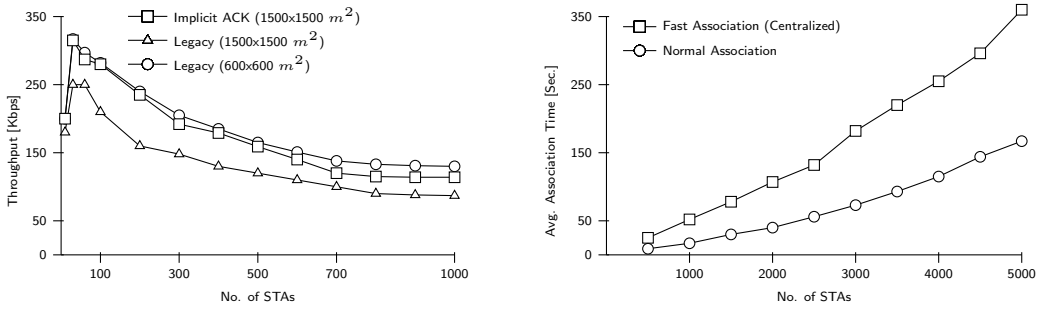


Figure 17: Throughput performance with hidden nodes considering (a) TIM STAs, (b) non-TIM, and Unscheduled STAs with 1000 STAs, 10 groups

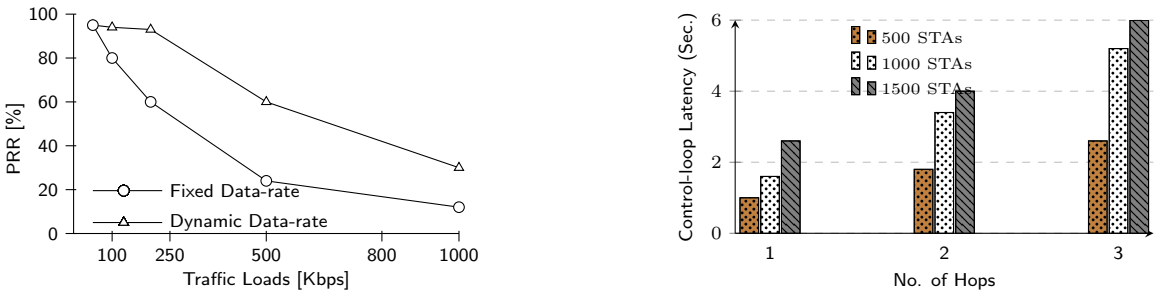
using the Transmission opportunity (TXOP) mechanism in the relay-based distributed network. Moreover, TXOP-based sectorization facilitates associating stations with different relay-enabled BSSs. Block ACK is suitable for better channel efficiency in such a multi-hop network. This scheme allows for sending multiple frames in a burst with a gap of Short Interframe Space (SIFS) time, which are acknowledged by a single BACK. An extra Block Request (BREQ) is sent to the sender in explicit block ACK. Considering implicit block ACK, we have analyzed a relay-based network as shown in Fig. 18a. Increasing the coverage range and number of STAs, the performance of legacy ACK and implicit block ACK is measured. It is clear that the block ACK based schemes improve performances. However, due to the hidden node problems in RID based communication, throughput performance decreases in the case of 1-hop results. Overall, the TXOP mechanism in the relay-based distributed network, coupled with sectorization and Block ACK schemes, plays a pivotal role in optimizing performance. Fig. 18b depicts different association time requirements by varying the number of STAs for different network sizes with group size=5 and RAW size= 15 slots.

IEEE 802.11ah supports multiple MCSs and channel bands for providing dynamic data rates. We simulate MCS0 with channel sizes 1 MHz and 2 MHz to see the performance of the network. As shown in Fig. 19a the PRR value is higher in the case of dynamic data-rate support employing dynamic MCS configuration. This configuration is orchestrated from the AP node, underlining the importance of dynamic MCS adaptation for improved network performance. For supporting sensing and actuation in an example Industry application, control-loop latency traffic management is an important performance metric. Keeping computation modules in AP and relays, different actuation policies can be executed at the run-time. Fig. 19b shows the control-loop, which is the time difference between sensing time and actuation. For a network with a higher number of nodes, the contention and RAW mechanism increase delay. Similarly, taking action from the nearest relay reduces the delay significantly. Similarly, when there are more nodes and multiple hops involved, the traffic load increases, resulting in decreased throughput. The increase in the number of nodes leads to higher time consumption, further diminishing the overall throughput performance.



(a) (b)

Figure 18: (a) Throughput achieved using legacy and implicit ACK mechanism in a relay based distributed network, (b) Association time required for different network sizes with group size=5 and RAW size= 15 slots



(a) (b)

Figure 19: (a) Average Packet Received Ratio (PRR), and (b) Average Control-loop latency

7.1. Discussion on the Performance of IEEE 802.11ah for NGI

The NGI strives to deliver interoperable communication capabilities for diverse objects connected to the Internet, focusing on scalability, interoperability, low latency, and extended coverage with lower power consumption. In our exploration, IEEE 802.11ah emerged as a promising solution, addressing challenges with its support for over 8000 devices in a single AP. Our performance analysis showcased its seamless association with up to 5000 devices, demonstrating a well-balanced throughput performance that can be replicated with a higher number of devices.

A pivotal aspect of NGI is the efficient forwarding of data through relays in multi-hop networking. Our analysis, extending to three hops, focused on control loop communication, revealing a delay suitable for automating diverse IoT applications, even in scenarios with considerable distances between sensor and actuator devices. The dynamic data rate support in IEEE 802.11ah proved instrumental in efficiently deploying IoT devices, considering critical parameters such as energy consumption and data rate requirements. This discussion seamlessly integrates with the overarching requirements for next-generation networking. Zooming into specific performance analyses, our

exploration of hidden nodes detailed their impact on throughput. As the coverage area expanded, we noted nuanced influences on different types of stations, emphasizing the interplay of network size and coverage.

Investigating the TXOP mechanism revealed substantial performance enhancements. Sectorization through TXOP facilitated efficient association of stations with relay-enabled Basic Service Sets (BSSs), further optimized by the application of Block ACK. Simulating MCS with varying channel sizes underscored the network's responsiveness to dynamic data-rate support. A notably higher PRR in dynamic MCS configuration, orchestrated from the AP node, highlighted its effectiveness. In addressing control-loop latency management, our analysis emphasized strategic computation module distribution in both the AP and relays. This approach enabled diverse actuation policies at runtime, with proximity to the nearest relay significantly reducing delay—an advantageous feature for efficient traffic management in industry applications.

The examination of network scale and multiple hops uncovered proportional increases in contention and delay. Conversely, a rise in the number of nodes led to heightened time consumption, emphasizing the need for a balanced network design to optimize throughput. This in-depth analysis contributes valuable insights to the evolving landscape of Next-Generation IoT, showcasing the robustness of IEEE 802.11ah in diverse scenarios. Future research directions will further refine and extend the applicability of this technology in NGI architectures.

8. Open Research Challenges and Future Direction

With the decisive engagement of IEEE 802.11ah technology in the application scenarios such as smart city, Industrial IoT, smart transportation, and name a few more for its significant features. This WiFi standard has some major contributions towards efficient channel access mechanism and energy consumption strategy, service to a wide area up to 1km distance with a single hop and also support more than eight thousand devices. And we embed this advanced technology for NGI Internet of Things architecture towards a more efficient, smart, and flexible IoT ecosystem. However, this technology is also facing some significant research challenges. In this section, we initially discuss the Current state of the development of IEEE 802.11ah hardware and software; thereafter, we endeavour to present some significant open research challenges and future directions concerning NGI architecture needs to be capable of self-configurable, self-manageable, and self-optimized.

8.1. Current state of the development of IEEE 802.11ah Hardware and Software

The IEEE 802.11ah standard serves a valuable purpose in both industry and research domains. While only a few organizations have successfully launched hardware or protocol stacks based on 802.11ah, Table 10 provides a comprehensive list of these organizations and highlights the features of their 802.11ah-based products. It is important to note that the specification of 802.11ah is still in the draft stage. Although numerous research papers delve into the

MAC and PHY layer concepts in detail, there are still many interesting issues that need to be considered, motivating the development of a smart and scalable system with a specific focus on MAC layer aspects. Notably, several hardware prototypes based on IEEE 802.11ah have already been introduced, showcasing the research excellence in this area [112, 113, 114, 115, 116].

The fabrication layout of an IEEE 802.11ah device encompasses multiple components and layers [117]. The RF front-end is responsible for signal transmission and reception, housing components like antennas, RF switches, power amplifiers, low-noise amplifiers, filters, and impedance matching networks. The transceiver, consisting of mixers, frequency synthesizers, modulators, demodulators, and signal conditioning circuits, handles data signals [88]. The baseband processor performs digital signal processing tasks, including modulation, demodulation, encoding, decoding, error correction, and channel access control, using DSPs, microcontrollers, and specialized digital circuits [118]. The device's memory component stores firmware, software, configuration data, and temporary operational data, comprising non-volatile memory (e.g., Flash memory) and volatile memory (e.g., RAM). The power management unit regulates power supply across different sections, featuring voltage regulators, power switches, and energy-saving capabilities for optimized power consumption. Various interfaces such as USB, UART, SPI, I2C, GPIO, and Ethernet enable external connections [88]. All these components are typically mounted on a PCB, which provides electrical connections, signal routing, and mechanical support for the device.

To facilitate research in this area, several simulation models have been developed using tools such as NS-3, Matlab, and OMNeT++ [119, 120, 121, 109].

8.2. Practical considerations for deploying IEEE 802.11ah in real-world IoT scenarios

While the potential of IEEE 802.11ah for Next-Generation IoT (NGI) is promising, the practical deployment of this technology involves several considerations, including cost and interoperability. The deployment of any new communication technology in IoT scenarios necessitates a thorough analysis of the associated costs. In the case of IEEE 802.11ah, factors such as hardware expenses, software development, and infrastructure upgrades must be considered. The cost-effectiveness of IEEE 802.11ah solutions should be evaluated concerning the scale of deployment and the specific requirements of NGI applications. Additionally, a comparison with alternative technologies should be made to assess the overall economic viability. Interoperability is a critical aspect of IoT ecosystems, where a multitude of devices from various manufacturers coexist. The successful integration of IEEE 802.11ah hinges on its ability to interoperate seamlessly with existing IoT devices and technologies. We recognize that interoperability challenges can arise due to differences in communication protocols, data formats, and security mechanisms. Addressing these challenges is imperative to ensure a smooth transition to IEEE 802.11ah and to foster a cohesive and interconnected NGI.

Table 10: Currently available 802.11ah-based hardware **and protocol stacks**

Product Name	**Manufacturer**	**Type of Product**	**Key features**
NRC7292	Newracom	STA	The NRC7292 is a compact Wi-Fi module with a built-in radio transceiver, combining application processing and networking capabilities for a powerful solution.
NRC7292 EVK	Newracom	Gateway, STA	Halo TD-XPAH offers long-range WiFi (1+ km), plug-and-play setup with built-in GUI firmware. Supports 902-928 MHz frequencies (US, Canada, Japan, Australia, NZ, South Korea). Includes antenna for easy installation.
M2B7111	Methods2Business	STA	M2B7111 ensures seamless compatibility with devices from different vendors, minimizing development risks.
M2B7211	Methods2Business	AP	Ideal as AP for home or small office
ORC8050 Wi-Fi 802.11ah	Orca systems	STA	ORC8050 is a global Wi-Fi 802.11ah product for long-range, low-power applications, supporting worldwide frequencies with 16MHz bandwidth.
Halo TD-XPAH	Teledatics Incorporated	STA	Halo TD-XPAH enables WiFi service beyond 1 km with plug-and-play setup via built-in GUI firmware. Supports 902-928 MHz frequencies in US, Canada, Japan, Australia, New Zealand, and South Korea. Includes antenna and pigtail for easy installation.
PCS2100	Palma Ceia SemiDesign	STA	PCS2100 supports 1 MHz and 2 MHz operation, with Wi-Fi extensions for IoT. Covers 755-928 MHz spectrum to meet regulations. Provides high receiver sensitivity, superior phase noise, and narrow-band transmission for reliable long-range connections.
PCS2500	Palma Ceia SemiDesign	AP	PCS2500 supports 1 MHz and 2 MHz operation with Wi-Fi extensions for IoT. Covers 755-928 MHz spectrum. Offers high sensitivity, low phase noise, and superior linearity for reliable long-range connections. Includes external FEM TR switch and PA (+26dBm power).
AP-100AH	Silex Technology	AP	AP-100AH access point provides a 3 km range and accommodates up to 675 clients. Seamlessly integrates 802.11ah into infrastructure, enabling communication with a RADIUS server for 802.1X authentication of client devices.
BR-100AH	Silex Technology	Bridge	BR-100AH enables secure long-range 802.11ah WLAN connectivity with high data throughput. Seamless integration for Ethernet devices without driver or application changes. Connects multiple devices to wireless LAN via a wired LAN port in Multi client mode.
SX-NEWAH	Silex Technology	STA	Operating Frequency: 902.0 928.0 MHz, Transmit Power: 23 dBM, Rx Min Sensitivity : -86 -106 dBM
AP7688-WHM	AsiaRF Co., Ltd	AP	Long-range transfer of Video and IoT, high data rate of up to 150Mbps, plug and play functionality, support for booting from flash, and maximum data rate of 32.8Mbps at 8MHz channel bandwidth.
AP7628-UHM	AsiaRF Co., Ltd	Gateway	IoT Gateway with Ethernet to USB C Dongle functionality, support for USB drivers on Windows, MAC, UEFI (PXE, UNDI), and Linux platforms.
MM6104 SoC	Morse Micro Ltd	SoC	Single chip solution integrates Radio, PHY, and MAC sections as per IEEE 802.11ah standard. Supports data rates up to 15 Mbps with programmable radio operation in 850MHz to 950MHz bands globally. Battery-friendly for battery-operated applications.
MM6108 SoC	Morse Micro Ltd	SoC	Single chip integrates IEEE 802.11ah Radio, PHY, and MAC sections. Supports data rates up to 32.5 Mbps. Programmable radio operates in 850MHz to 950MHz bands. Flexible RF interface. High linearity receiver, battery-friendly.
Aegis-IP 802.11 ah	Aegis-IP	SoC, STA, Router, AP	STA Infrastructure BSS supports 1MHz and 2MHz channel bandwidth. Single stream transmission with various frame types. MCS options for 0, 1, 2, and 10. Includes Normal Guard Interval, Fixed Pilot, and QoS. Supports Block Acknowledgement, TWT, RAW, and Authentication control. Provides up to 2 Mbps performance with 1 km range.

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While the potential of IEEE 802.11ah for Next-Generation IoT (NGI) is promising, the practical deployment of this technology involves several considerations, including cost and interoperability. The deployment of any new communication technology in IoT scenarios necessitates a thorough analysis of the associated costs. In the case of IEEE 802.11ah, factors such as hardware expenses, software development, and infrastructure upgrades must be considered. The cost-effectiveness of IEEE 802.11ah solutions should be evaluated concerning the scale of deployment and the specific requirements of NGI applications. Additionally, a comparison with alternative technologies should be made to assess the overall economic viability. Interoperability is a critical aspect of IoT ecosystems, where a multitude of devices from various manufacturers coexist. The successful integration of IEEE 802.11ah hinges on its ability to interoperate seamlessly with existing IoT devices and technologies. We recognize that interoperability challenges can arise due to differences in communication protocols, data formats, and security mechanisms. Addressing these challenges is imperative to ensure a smooth transition to IEEE 802.11ah and to foster a cohesive and interconnected NGI.

8.4. Research scopes and future directions

The 802.11ah standard supports the PHY and MAC layers but lacks native support for upper layers like TCP/IP. To overcome this, a new protocol or adaptation layers needs to be developed for seamless integration with the existing internet system. Additionally, leveraging SDN, edge computing, and machine learning can further enhance the network's capabilities. This combined approach enables efficient resource management, intelligent decision-making, and improved performance, unlocking the full potential of 802.11ah for innovative applications. A few future research directions are elaborated in the following section. In the remainder of this section, we will delve into some of the key research challenges and discuss future directions that need to be considered alongside the implementation of the 802.11ah standard for a NGI infrastructure.

Network Management: For a flexible and reliable network functionality, we must be aware of the network's global state to manage the network efficiently. But acquiring the global state is itself a difficult task in traditional architecture. We may prioritize the flow of packets, handle loads on the channel, and many more when we have access to network management remotely. We may also require reconfiguring network functionalities as user or situation demands. Software Defined networks (SDN) is a solution for network management and reconfigurability.

Time Sensitive Applications:: The NGI must be equipped with functionalities to serve time-sensitive applications such as Healthcare and Auto-driving vehicular networks. In such a case, a delay in seconds may take lives away or lead to massive destruction. IEEE 802.11ah is still facing challenges to optimize delays up to the mark.

Network Slicing: Recently, network slicing in Radio Access Technologies (RATs) has been proposed to divide the wireless network infrastructure into isolated logical slices for provisioning dedicated QoS [122]. It is expected

that in such a complex network architecture such as NGI, slicing will significantly impact supporting heterogeneous requirements of applications. Due to the availability of multiple configurations as controlled by the AP using DTIM, TIM, etc., IEEE 802.11ah is a suitable candidate technology. However, making an IEEE 802.11ah network complete programmable is still an issue.

Softwarized Network Control: IEEE 802.11ah enabling technology serves as a communication strategy to a large area with thousands of IoT devices connected with multi-hop facilities. A large number of links are available to connect the devices to the network. These links cover a wide area, and they should be capable of self-repairing for any link disturbance. Embedding SDN technology is a reliable solution for network flexibility, management, and for issues mentioned above [123]. In SDN-enabled IoT network architecture, the advantages we taste are flexibility and resiliency for unnecessary links available. We opt for SDN for a scalable heterogeneous network for better handoff, device association, and resource allocation for various healthcare, Industry, and Smart Grid applications. Again, for a large scale flexible and virtual network configuration in the network, devices with OpenFlow [124] and P4 [125] based API can be used widely through programmable and configurable AP and relay devices.

Intend Driven Networking: Intent-driven network (IDN) is defined as a self-driving network that uses decoupling network control logic and closed-loop orchestration techniques to automate application intents [126]. In the IDN, the intent needs to specify the need of the network and configure/reconfigure/program accordingly to get serviced the new policies based on the intend specified for the network's execution. With the enhancement of IDN, we can move a step ahead towards the automation technologies in services such as security [127], resource management [128], and transformations, e.g. Natural language into network strategies [129]. However, a few more extensive researches are done in literature [127, 130, 131, 128, 132]. Still, a more detailed analysis is needed for implementation purposes focusing on long-distance communication, energy-optimized service, and secure network strategies incorporated with NGI.

QoS: In latency-sensitive IoT applications, the packet must be transmitted within a fixed time. To transmit such kind of packets, the network should felicitate smooth transactions with some reserved bandwidth. In a huge IoT network scenario with thousands of devices associated, there may be some devices that do not get a chance to transmit as frequently as their need. A detailed performance analysis of real-time VoIP service has been carried out in [133].

Security: Along with the innovative MAC and PHY layer solutions for NGI, IEEE 802.11ah also creates many security challenges due to its association, extensive coverage, and relay features. The remote applications and coverage make the communication more vulnerable to attacks. Also, identifying the relay's owner and authorization of using the relay to pass traffic through, etc., are also some of the issues that need to be addressed. Moreover, designing a lightweight security solution for resource-constraint NGI is a challenge.

Edge Intelligence: IEEE 802.11ah-based wireless network for NGI will facilitate a wide coverage having multiples APs and relays. For provisioning low latency actuation and analytic in application scenarios such as industry and healthcare. The use of edge intelligence mechanisms such APs and relays will have considerable potential for improving network's and application's requirements. Instead of using cloud analytic and cloud intelligence [134], edge intelligence will enable localized analytic and decisions.

Coexistence: Although larger overage and sub-1 GHz reduce the coexistence issues significantly, the use of multiple APs and relays with the same or nearer frequency band will still have the interference issues [135]. Moreover, other technologies such as IEEE 802.15.4 also have sub-1GHz channel configuration, which is supposed to be deployed alongside 802.11ah [136]. Therefore, interference due to coexistence is still an issue.

Mobile IoT and Augmented Reality: With the increasing number of mobile IoT and Augmented Reality (AR) applications in different domains such as health, education, etc., issues such as security, performance, reliability, and storage limitations are also increasing [137]. IEEE 802.11ah can connect such devices for IoT services. But, how dynamic characteristics of devices can influence the performances that need to see.

8.5. Limitations and Constraints

While IEEE 802.11ah presents a promising solution for Next-Generation IoT (NGI), it is imperative to acknowledge certain limitations and constraints that may impact its widespread adoption and application. One of the inherent constraints of IEEE 802.11ah is its reliance on sub-1 GHz frequencies for extended coverage. While this allows for communication over larger distances, there are limitations in achieving high data rates over such ranges. The trade-off between coverage and data rate is a critical consideration, and applications requiring both extensive coverage and high data transfer speeds may face challenges in finding an optimal solution within the IEEE 802.11ah framework. In scenarios where multiple IEEE 802.11ah networks coexist or overlap with other wireless technologies, interference issues may arise. Despite efforts to reduce interference through sub-1 GHz frequencies and efficient channel access mechanisms, practical challenges related to coexistence with other wireless standards, such as IEEE 802.15.4, still need careful consideration. While IEEE 802.11ah incorporates security mechanisms, the extensive coverage and relay features introduce unique challenges. Ensuring secure communication over long distances and through relays poses additional complexities. The design of lightweight and robust security solutions for resource-constrained NGI environments is an ongoing research challenge.

9. Conclusions

This paper presented a Next-Generation IoT architecture based on features of IEEE 802.11ah. To satisfy the requirements – scalability, energy-efficiency, complexity, interoperability, etc., IEEE 802.11ah proposes potential

solutions suitable for NGI. We discuss the key requirements of an NGI and study the feasibility of existing communication technologies with them. IEEE 802.11ah is a suitable solution for link-layer connectivity due to the support for more extensive coverage, low power consumption, dynamic data rate, etc. This paper discusses the components and protocol stack of an NGI in detail. Finally, we highlight a few future research directions and challenges.

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